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Genome-wide identification, evolution, and expression analysis of *StNRT* (nitrate and peptide transporter) gene family in potato (*Solanum tuberosum* L.)

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Abstract

Background Nitrate and peptide transporters play crucial roles in the uptake, and allocation of nitrate in plant cells and tissues. They are essential in the utilization of nitrogen, as well as in plant development, crop production and yield formation. The NRT family (Nitrate Transporter/Peptide Transporter family, NRT/NPF) is one of the largest transporter families in the plant kingdom. Some family members have been extensively studied in thale cress (*Arabidopsis thaliana*) and rice (*Oryza sativa*). Although it has been reported in potatoes, it has not been comprehensively and systematically analysed yet. The purposes of this research were to comprehensively identify the NRT genes in the potato genome, to systematically analyze the gene structure, chromosomal distribution, and *cis-elements* in promoter regions, to identify protein conserved domains and motifs, and to study the expression profile of the *StNRT* gene in different tissues and under nitrogen stress.

Results Bioinformatics and transcriptome analyses of the *StNRT* gene family were conducted to dissect the structure, evolution, and expression of the *StNRTs* in *Solanum tuberosum* L. In total, 81 *StNRT* genes were identified and classified into 4 clades. Chromosomal localization analysis revealed that the 81 *StNRT* genes were located on 12 chromosomes of potato. Among these genes, 2 pairs of genes (*StNRT53/56* and *StNRT61/62*) were predicted to be tandemly duplicated genes, and 8 pairs of genes were segmentally duplicated genes. The collinearity analysis revealed that the *StNRTs* in potato were closely related to the *SINRTs* in tomato. The expression profiles of the *StNRTs* in different tissues and under different nitrogen conditions revealed distinct spatial expression patterns of these genes and their potential roles in response to various nitrogen conditions. Multiple *StNRT* genes exhibited differential expression in the roots, stems, and leaves of Qingshu 9 and Xiazhai 65 potato seedlings, indicating their potential key roles in regulating nitrate uptake in potato.

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Conclusions This study systematically characterizes the gene structure, conserved protein domains and motifs, evolutionary relationships, and expression profiles of the StNRT gene family in *S. tuberosum*. These findings provide critical insights into the functional mechanisms of StNRTs, and identify candidate genes for improving the nitrogen use efficiency (NUE) in potato cultivation.

Keywords Potato, Nitrate transporter, Peptide transporter, Nitrogen stress, Gene family function analysis, Gene expression pattern

Introduction

Nitrogen is an essential element for nucleotide, amino acid, chlorophyll, and alkaloid metabolites [1]. It plays an important role in the construction of plant cells, and tissues, and in yield formation and is also the primary factor limiting plant growth and yield [2]. Potato is an important noncereal food crop and a highly heterozygous autotetraploid species [3, 4]. In China, the planting area is 4.67 million hectares per year, with an annual output of approximately 90 million tons. In actual production, the nitrogen use efficiency (NUE) is less than 40% [5]. Cultivating varieties with high nitrogen use efficiency is an important way to improve nitrogen use efficiency, including nitrogen absorption, assimilation, and reallocation processes [2]. Nitrate is the main form of nitrogen absorbed by potato, which is taken up and transported by nitrate transporters in the roots. The NRT family (Nitrate Transporter and Peptide transporter family, NRT/NPF) plays an important role in the absorption, transport and distribution of nitrate nitrogen in plant cells and tissues [6]. It is one of the largest transporter families in the plant kingdom [7] and plays a crucial role in nitrogen use efficiency and crop production.

Various studies have focused on characterizing and cloning different members of this gene family in various plant species. In *Arabidopsis*, the *NRT1.8* gene functions in nitrate removal from xylem sap and mediates cadmium tolerance [8]. Additionally, the *NRT2.4* gene in *Arabidopsis* plays a dual role in the roots and shoots of nitrogen-starved plants [9]. The *NRT2* gene family has been characterized in *Arabidopsis*, with studies showing interactions between different components of the nitrate uptake system. A total of fifty-three *NPF* genes, seven *NRT2* genes, and two *NAR2* genes were identified in *Arabidopsis Thaliana* [6, 10]. *AtNRT1.1* is the first nitrate transporter identified and cloned in plants. It has the functions in nitrate absorption and transport, and auxin transport and regulates nitrate-mediated root development. *AtNRT1.1* can respond to changes in the environmental nitrate concentration by changing its level of phosphorylation to regulate nitrate transport activity [11], and it is a receptor that senses nitrate signals [12]. Most *NPFs* have low nitrate affinity, whereas *AtNRT1.1* and *OsNRT1.1B* have dual affinities [13]. In rice, 93 nitrate transporters were identified, including 70 *NPFs*. The *OsNRT1.1A* gene, a member of the nitrate

transporter 1/peptide transporter family, has been identified as a key regulator of nitrogen utilization and flowering, leading to high yield and early maturation [14]. *NRT1.1B* participates in regulating the establishment of root microbial communities and promotes the mineralization of organic nitrogen, thereby mediating the interaction between plants and microorganisms. The rhizosphere of indica rice is richer in microbial communities involved in nitrogen metabolism than that of japonica rice is, resulting in higher nitrogen use efficiency in indica rice varieties than in japonica rice [15, 16]. The natural variation of the *NRT1.1B* protein is that the 327th amino acid of indica rice is methionine (Met), whereas that of japonica rice is threonine (Thr). The *NPF* family facilitates the transport of a multiple substrates, encompassing phytohormones such as abscisic acid (ABA) and gibberellins (GAs), as well as metabolites including jasmonic acid-isoleucine (JA-Ile), dipeptides, and polypeptides. *OsNPF3.1* is an important low-affinity nitrate and plant hormone transporter gene, that is crucial for rice tillering and nitrogen use efficiency. This gene is strongly induced by nitrate, ABA and gibberellin 3 (GA3) in the roots, stems and leaves of rice and has a significant effect on rice tillering and nitrogen use efficiency. The expression of the *OsNPF3.1* gene is positively correlated with the number of tillers. Its overexpression significantly promotes the accumulation of ABA in the roots and GA at the base of the plant, thereby inhibiting the growth of lateral buds and rice tillering, especially under high nitrate concentrations [17, 18]. Experiments have shown that *OsNPF3.1* is a pH-dependent low-affinity nitrate transporter and is also a transporter of ABA and GA3. Most *NRT2s* are high-affinity nitrate transporters [6], and the high-affinity transport system is activated at low nitrate concentrations. *NAR2* interacts with *AtNRT2.1*, *AtNRT2.2*, *AtNRT2.5*, *OsNRT2.1* and *OsNRT2.3a* on the cell membrane to improve nitrate transport efficiency [6, 19]. The overexpression of *OsNAR2.1* can significantly improve nitrogen absorption and drought resistance in rice, whereas *OsNAR2.1* knockout plants have the opposite effect [20]. Studies in tomato have shown that the expression of a cluster of genes encoding *NPF* members is downregulated in a mutant with a low-SGA phenotype, indicating the importance of these genes in plant growth and development [21]. Furthermore, 77 *NPFs*, 7 *NRT2s*, and 1 *NAR2* were found in *Solanum lycopersicum* [22].

Overall, a comprehensive understanding the nitrate and peptide transporter (NRT/NPF) gene family is crucial for elucidating nitrogen metabolism and stress responses in various crops species, including potato. While Zhang et al. [23] identified 39 *StNRT* genes using PGSC DM v4.03 genome, our study significantly expands this number to 81 leveraging the SolTub_3.0 genome. The objectives of this research were (i) to conduct a more comprehensive identification of *NRT* genes in the potato genome; (ii) to systematically analyze gene structure, chromosomal distribution, promoter sequences, conserved protein domains and motifs; (iii) to perform a phylogenetic analysis of *StNRT* genes within potato species; (iv) to compare the homology and phylogenetic relationships of potato *StNRT* proteins with NRT proteins from other species; (v) to investigate the tissue-specific expression patterns of *StNRT* genes in potato; and (vi) to examine the expression dynamics of *StNRT* genes in response to nitrogen stress. To date, no comprehensive whole-genome analysis of the *NRT* gene family in potato has been reported. In this study, we used whole-genome analysis to identify members of the *StNRT* gene family. Through bioinformatics and expression analysis, our findings provide a robust theoretical foundation for further research on potato nitrate transporter genes and pave the way for functional studies of these genes. Further investigations into the regulatory mechanisms and functional roles of these genes will offer valuable insights into improving crop yield and nutrient utilization efficiency.

Materials and methods

Genome-wide identification of potato *StNRT* genes

The potato gene and protein sequences referenced in this study were homozygous diploid SolTub_3.0 assembly for *Solanum tuberosum* (GCA_000226075.1) sequencing and the assembly of the genomic data of the DM 1–3 516 R44 cultivar (Potato Genome Sequencing Consortium) [24]. The *StNRT* genes were identified through a combining of protein sequence alignment using BLASTP (v2.12.0, with an e-value cutoff $\leq 1e-50$, query coverage $\geq 70\%$, identity $\geq 35\%$) and the Hidden Markov Model HMM search (hmmsearch v3.2.1, with pfam domains PF00854/PF07690 and an e-value cutoff $\leq 1e-5$). In the BLASTP approach, the NRT sequences were queried on the TAIR website, 41 genes were obtained. After screening, 37 *AtNRT* gene IDs were selected, and their corresponding protein sequences were downloaded based on the Aroport11 protein database. Using the Blast function of TBtools-II v2.136 [25], and 107 potato NRT protein IDs were identified. For the HMM method, the protein domains of the NRT/NPF gene family (with PF00854 and PF07690) were retrieved from the Pfam database [26], and 419 protein IDs were obtained using the simple HMM search function in TBtools-II. The

intersection of the 107 protein IDs from BLASTP and the 419 protein IDs from HMM search resulted in 107 overlapping proteins. After further screening to retain the longest transcript of each gene, a total of 81 *StNRT* genes were ultimately identified. The coding sequences (CDS) and protein sequences of the 81 *StNRT* genes were extracted using the Fasta Extract function in TBtools-II. Subsequently, the pfam domain and conserved protein domains of each gene family member were annotated through comprehensive analyses using InterProScan (<https://www.ebi.ac.uk/interpro/>), SMART (<https://smart.embl.de/>) [27], and UniProt (<https://www.uniprot.org/>) databases. Additionally, these analyses were further supported by the latest potato genome assembly data, DM8.1 (<http://www.bioinformatics-lab.cn/pubs/dm8/>) [28], and validated through blastn (v2.12.0, with an e-value cutoff $\leq 1e-10$, query coverage $\geq 90\%$, identity $\geq 73\%$) and Blastp (v2.12.0, with an e-value cutoff $\leq 1e-30$, query coverage $\geq 90\%$, identity $\geq 80\%$) to reconcile information for each family member. The same methodology was applied to the DM8 genome, resulting in the identification of a total of 96 genes.

Gene chromosome mapping

Utilizing the reference genome data, the chromosomal lengths and mapping data of *StNRT* gene family members were obtained. To visualize the chromosomal distribution, a map was generated using the MG2C (http://mg2c.iask.in/mg2c_v2.1/) online tool [29], which clearly depicts the localization of gene family members across the potato chromosomes.

Analysis of gene structure and Cis-elements in the promoter region

A comprehensive analysis of 81 *StNRT* genes was performed, including the extraction of their coding sequences (CDS) lengths, UTR lengths, introns, exons and chromosomal distribution information using TBtools-II software. The annotation file SolTub_3.0.57.gff3 was processed to extract the 1000 bps sequence upstream of the CDS for each *StNRT* gene and assumed it to be the promoter region. *Cis-elements* within those promoter sequences were identified using the PlantCARE database [30], with common *cis-elements* in promoter and enhancer regions and those lacking functional annotations subsequently filtered out. The gene structures, including UTRs, introns, exons, and predicted *cis-elements*, were visualized based on their respective genomic locations.

Conserved protein domains and motifs

To analyze conserved motifs within the potato *StNRT* protein sequences, the MEME Suite web server [31] (<https://meme-suite.org/meme/>) was utilized with a

specific algorithm. This analysis identified the top 20 conserved motifs, and the corresponding meme.xml files were generated for further use. Conserved domains within the protein sequences were identified using InterProScan (<https://www.ebi.ac.uk/interpro/>) and the Batch CD-Search tool [32] available on the NCBI website (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>). The identified motifs and conserved domains were then visualized using the Gene Structure View function in TBtools software [25], providing a comprehensive overview of the protein structural features.

Physicochemical properties, subcellular localization, transmembrane domain prediction and protein–protein interaction prediction

The physicochemical properties of the gene family members' proteins, including amino acid number, molecular formula, molecular weight, isoelectric point, instability index, and aliphatic index, were determined using the SIB Swiss Institute of Bioinformatics ExPASy website (<https://web.expasy.org/protparam/>). The transmembrane helices (TMHs) of the proteins were predicted using the TMHMM-2.0 online tool (<https://services.healthtech.dtu.dk/services/TMHMM-2.0/>) [33]. Subcellular localization predictions were conducted using the WoLF PSORT online tool (<https://wolfsort.hgc.jp/>) [34]. To explore the functional relationships among the target genes, protein–protein interaction (PPI) (Score > 0.95) analyses were performed using the STRING database (<https://string-db.org/>).

Phylogenetic and collinearity analysis

Reference genome sequences and annotation files for *Solanum lycopersicum* (SL3.0.dna.toplevel.fa and SL3.0.57.gff3) [35], *Nicotiana attenuata* (NIATTr2.dna.toplevel.fa and NIATTr2.57.gff3) [36], *Arabidopsis thaliana* (TAIR10.dna.toplevel.fa and TAIR10.57.gff3) [37, 38], and *Oryza sativa* Japonica group (IRGSP-1.0.dna.toplevel.fa and IRGSP-1.0.57.gff3) [39] were downloaded from the Ensembl database. The downloaded files included the top-level genome assemblies and their corresponding GFF3 annotation files. Collinearity analysis and visualization within and between species were performed via the OneStepMCScanX (e-value $\leq 1e-10$) and Dual Synteny Plot functions of TBtools software, and collinearity of *StNRT* genes was annotated. Through BLASTP (e-value $\leq 1e-10$) and literature searches, the protein sequences of the *NRT* gene family members of *S. lycopersicum* and *A. thaliana* were obtained. With Mega 11 software [40], the *NRT* protein sequence was uploaded for multi-sequence alignment, and aligned by applying the MUSCLE algorithm via the maximum likelihood method (ML). The number of bootstrap replications was 1000, Newick files were obtained, and the

online analysis was performed on the iTOL website [41] (<https://itol.embl.de/>) to construct the phylogenetic tree.

Gene expression pattern analysis

DM 1–3 516 R44—Gene Expression Matrix (TPM)—v6.1 was downloaded from the SpudDB (https://spuddb.uga.edu/dm_v6_1_download.shtml) website [42], the expression data of 81 *StNRT* genes were extracted, the expression patterns of 81 *StNRTs* gene family members were analyzed, and a gene expression heat map was drawn via TBtools software.

Potato nitrogen treatment, RNA extraction and transcriptome sequencing, protein extraction and proteomic sequencing

Qingshu 9 (*S. tuberosum* cv. 'Qingshu 9') and Xiazhai 65 (*S. tuberosum* cv. 'Xiazhai 65') [43], are two dominant potato cultivars in Qinghai Province. Previous studies conducted by our research group evaluated the nitrogen use efficiency (NUE) of 32 potato varieties under the field conditions, revealing that Qingshu 9 exhibits high nitrogen use efficiency, while Xiazhai 65 demonstrates tolerance to low nitrogen conditions. Both cultivars were provided by the Qinghai Academy of Agriculture and Forestry Sciences. Tissue culture seedlings of the two potato varieties were grown under three nitrogen conditions—high nitrogen (A, 40 mM), medium nitrogen (B, 8 mM), and low nitrogen (C, 2 mM)—for 30 days in a controlled environment with a 16/8 h light/dark cycle, 1200 lx light intensity, and a temperature of 22 ± 2 °C at the Qinghai Academy of Agriculture and Forestry Sciences. RNA-Seq and LC–MS analyses were performed by Shanghai Bioprofile Biotechnology Co., Ltd (Shanghai, China). Transcriptome and proteome sequencing were performed on the roots, stems, and leaves of the two varieties under three nitrogen levels, with three biological replicates for each sample.

Total RNA was isolated using TRIzol reagent, and RNA concentration and purity were measured using a Nano-drop spectrophotometer. RNA integrity was verified through RNA-specific agarose electrophoresis. Library quality was assessed using an Agilent 2100 Bioanalyzer (Agilent Technologies Inc., California, USA) and an Agilent High Sensitivity DNA Kit (Agilent Technologies Inc., California, USA, 5067–4626). Protein extraction was performed using SDT lysis buffer (4% SDS, 100 mM Tris–HCl, pH 8.0). LC–MS/MS analysis was carried out on an Orbitrap Astral mass spectrometer coupled with a Vanquish Neo UHPLC system (Thermo Fisher Scientific). The DIA MS data were analyzed using DIA-NN 1.8.1, and the MS data were searched against the UniProtKB reviewed (Swiss-Prot) database.

Real-time quantitative PCR validation

RT-qPCR analysis was performed with a LightCycler 480II instrument (Roche). In triplicate, StGAPDH was used as the internal standard according to the following method: 1 µg of total RNA from the same batch of RNA for high-throughput RNA-Seq was used for first-strand cDNA synthesis via the PrimeScript™ 1st Stand cDNA Synthesi Kit according to the supplier’s protocol. PCR was conducted in a total volume of 20 µL with 1 µL of cDNA template, 400 nM forward primer, 400 nM reverse primer, and 10 µL of 2× SYBR real-time PCR premixture. Quantification was performed with Bio-Rad CFX manager software (V3.1). Quantitative verification was performed via a relative quantitative method ($2^{-\Delta\Delta Ct}$).

Gene expression and function analysis

HTSeq (v0.9.1) was used to statistically align the read count value of each gene to the original expression level of the gene. To make the gene expression levels of different genes and different samples comparable, the expres-sion level was normalized by FPKM, where FPKM is the number of fragments per kilobase length of a gene per million fragments. For pair-end sequencing, each frag-ment has two reads, and FPKM counts only the number of fragments that can be compared to the same transcript by two reads. Differential expression analysis between the two comparison combinations was performed via DESeq software (v1.20.0). DEGs were identified using DESeq2 with thresholds of $|\log_2FC| > 1$ and Benjamini–Hochberg FDR-adjusted $P < 0.05$. Raw P-values were adjusted for multiple testing using the default DESeq2 pipeline. For GO enrichment analysis, all genes were annotated with GO terms, and the number of DEGs associated with each term was calculated. The TopGO package was used to perform GO enrichment analysis on the DEGs, the P value was calculated via the hypergeometric distribu-tion method (the standard of significant enrichment was

a P value < 0.05), and the GO term associated with sig-nificantly enriched DEGs were identified to determine the main biological functions associated with the DEGs. Heatmaps of transcriptomic data were generated based on log2-transformed FPKM values from RNA-seq anal-ysis. Similarly, proteomic heatmaps were constructed using log2-transformed LFQ (label-free quantification) intensities derived from mass spectrometry analysis. Visualization of heatmaps was performed using TBtools software.

Results

Genome-wide identification and analysis of the StNRT gene in potato

Through BLASTP analysis, 107 proteins were identified, while 419 proteins were obtained via HMM search. The intersection of these two methods yielded 107 proteins. To ensure gene uniqueness, multiple transcripts of the same gene were screened, and only the longest transcript was retained. The Pfam and InterPro accession numbers of each family member were verified using the SMART and UniProt databases. Ultimately, 81 StNRT gene mem-bers were identified on the potato Soltub_3.0 genome (Additional file 1 Table S1 and Additional file 2). These genes were systematically named StNRT01 to StNRT81 according to their chromosomal positions (Chr_1 to Chr_12).

Chromosome mapping and the number of *StNRT* genes on each chromosome are as follows (Fig. 1, Table S1): 81 *StNRT* genes are mapped on 12 chromosomes, with Chr_6 having the most mapped to 15 genes, followed by Chr_5, which maps to 13 *StNRT* genes. Chr_3, 11 and 12 mapped to 7 genes.

The twelve chromosomes, designated Chr_1 to Chr_12, are shown with the number of *StNRT* genes identified on each chromosome indicated above the correspond-ing chromosome. The *StNRT* genes were systematically

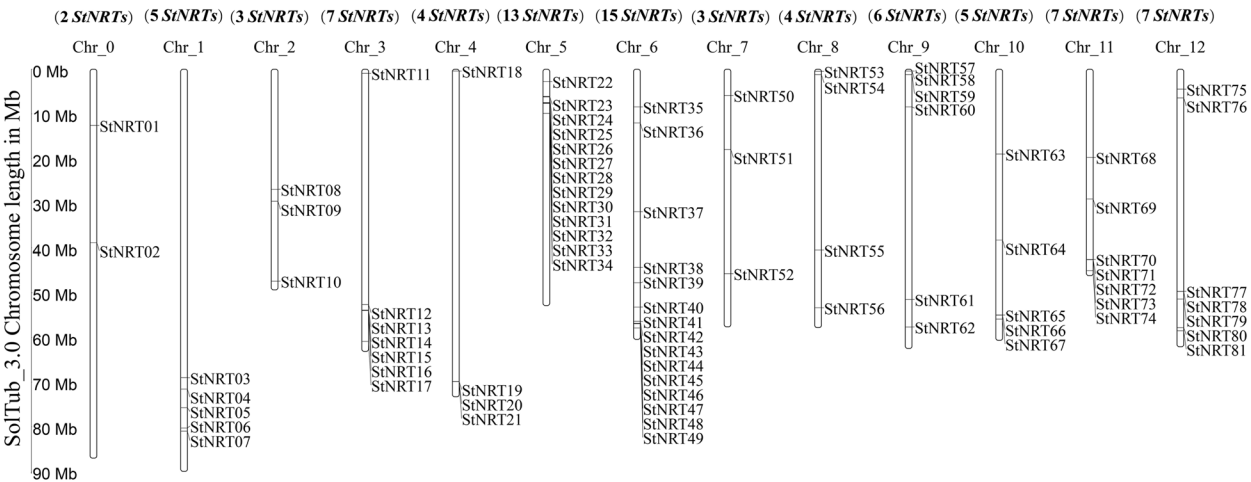


Fig. 1 Physical map of the *StNRT* genes on the SolTub_3.0 genome of potato (*Solanum tuberosum*)

named from *StNRT01* to *StNRT81* based on their chromosomal positions. Notably, *StNRT01* and *StNRT02* have been mapped to SolTub_3.0 genome Chr_0 (a pseudochromosome containing unanchored scaffold sequences) due to limitations in gene annotation. However, analysis of the latest potato T2T DM8.1 genome revealed that *StNRT01* was located on DM8.1_chr04, while *StNRT02* was mapped to DM8.1_chr06 (Fig. S1). The scale bar on the left indicates length of the SolTub_3.0 genome chromosomes in megabases (Mb). The physical map was generated using MG2C_v2.1.

Blastp analysis was performed using the latest DM8 genome, resulting in the identification of 96 *StNRT* genes on the T2T genome (Table S2). Among these, the highest number of genes (18) were located on DM8.1_chr06, followed by DM8.1_chr05, which harbored 15 genes. Notably, two genes could not be mapped to specific chromosomes in the SolTub_3.0 genome assembly. However, on the DM8.1 assembled genome (Additional file 3 Fig. S1), DM8C04G00100.1 (*StNRT01*) was mapped to DM8.1_chr04, and DM8C06G30400.1 (*StNRT02*) was mapped to DM8.1_chr06. Additionally, DM8sca36G00050.1 (*StNRT52*), which was mapped to an unknown chromosome in the DM8.1 genome, was located on SolTub_3.0_Ch07.

Protein sequences were annotated using InterPro and visualized using TBtools (Fig. S2 and Fig. S3), revealing that the highest number of domains were identified for Phobius, TMhelix, and TRANSMEMBRANE. InterProScan analysis of the 81 *StNRTs* showed that 72 genes contained the POT family (PF00854) domain, 9 genes contained MFS_NRT2_like (cd17341) domain, and 8 of these also contained the Major Facilitator Superfamily (PF07690) domain. Additionally, One gene contained both High-affinity nitrate transporter accessory (PF16974) and NAR2 (PIRSF012939) domain. Since different transcripts of the same gene were eliminated when the SolTub_3.0 genome was used as the reference, two genes in the DM8 genome were identified to contain both the High-affinity nitrate transporter accessory and the NAR2 domain.

Physicochemical properties, phylogenetic analysis and conserved domains of *StNRT* proteins

The physicochemical properties of *StNRT* proteins were summarized in Table 1 and Table S3. The amino acid length of the *StNRT* protein ranged from 208 to 653 aa, with a mean and median length of 544 aa and 579 aa, respectively. Among these, 58 proteins were 500~600 aa in length, 14 proteins were less than 500 aa, and 9 proteins exceeded 600 aa. The mean and median molecular weights of the *StNRT* protein were 60.35 kDa and 64.28 kDa, respectively. *StNRT02* protein exhibited the lowest molecular weight (23.25 kDa), while *StNRT68* had

the highest molecular weight (71.90 kDa). The theoretical pI values ranged from 5.56 to 9.48, with an average of 8.357 and a median of 8.84. Based on the instability index (23.9~47.94, average 35.58), 61 proteins were predicted to be stable protein (instability index < 40), 20 proteins were unstable (instability index ≥ 40). Subcellular localization analysis revealed that most *StNRT* proteins are localized to the plasma membrane, while *StNRT46* and *StNRT48* are localized to the vacuolar membrane. Transmembrane helices (TMHs) analysis showed that 21 genes contained 12 TMHs, 20 genes contained 11 TMHs, 20 genes contained 10 TMHs, and 9 genes contained 9 TMHs. Notably, 72 genes contained 9 or more TMHs, while *StNRT12* had only 2 transmembrane domains. The aliphatic index ranged from 82.78 to 110.22 (average 99.752), indicating a high content of aliphatic amino acid in *StNRT* proteins, which correlates with good solubility and stability. The Grand average of hydropathicity (GRAVY) ranged from 0.042 to 0.696 (average 0.313), with *StNRT12* being the only protein with a negative GRAVY value (−0.257). Most *StNRT* proteins were classified as hydrophobic proteins. Amino acid composition analysis revealed that the predominant amino acids in *StNRT* proteins were leucine (Leu), serine (Ser), isoleucine (Ile), alanine (Ala), and glycine (Gly) (Table S3).

The phylogenetic tree and conserved domains/motifs of *StNRT* proteins were presented in Fig. 2 and Table S4. Phylogenetic analysis classified the *StNRT* proteins into four clades, designated C1 through C4. The C2 clade comprised 9 high-affinity nitrate transporter proteins, characterized by the major conserved domain was PLN00028 (nitrate transmembrane transporter), which belongs to the MFS_1 (Major Facilitator Superfamily, pfam07690) superfamily. Additionally, these proteins contained either the NarK (nitrate transporter NarK, COG2223, IPR004737) or MFS_NRT2_like (cd17341) conserved domains. In C1, C3 and C4 clades, all *StNRT* proteins contained the conserved domain PTR2 (POT family, pfam00854), except for *StNRT12*, which uniquely contained the NAR2 (high-affinity nitrate transporter accessory, pfam16974) domain. Different clades exhibited distinct conserved domains: C1 clade proteins contained the MFS_NPF1_2 (cd17416) and MFS_NPF3 (cd17415) domains; C3 clade proteins contained the MFS_NPF4 (cd17414) and MFS_NPF6 (cd17413) domains; C4 clade proteins contained the MFS_NPF5 (cd17417), MFS_NPF7 (cd17419), and MFS_NPF8 (cd17418) domains.

The distribution and sequence information of the 20 conserved motifs were illustrated in Fig. 2C/D and detailed in Table S4). Among the *StNRT* proteins, 71 proteins in C1, C3, and C4 clades contained Motif 3, while 70 proteins in these clades consistently contained Motifs 2, 9, 11, and 13 (Fig. 2D, Table S4). In C2 clade, which comprises 9 high-affinity nitrate transporter proteins,

Table 1 List of *StNRT* gene members

Gene No	Reference gene ID	Chr location	Cds length/bps	exons	Protein length/aa	TMHs	UniProt	Protein Name
<i>StNRT01</i>	PGSC0003DMG402001926	00:12,386,709:12,391,564	1722	4	573	11	M0ZP29	NPF1.2
<i>StNRT02</i>	PGSC0003DMG400022107	00:38,273,536:38,274,380	627	1	208	5	M1BZY4	NPF2.7
<i>StNRT03</i>	PGSC0003DMG400001671	1:67,991,306:67,996,008	1791	5	596	12	M0ZN17	NPF7.3
<i>StNRT04</i>	PGSC0003DMG400025961	1:70,501,630:70,503,321	1392	2	463	9	M1CG61	NPF7.3
<i>StNRT05</i>	PGSC0003DMG402027501	1:74,630,384:74,633,515	1728	5	575	11	M1CML8	NPF6.2
<i>StNRT06</i>	PGSC0003DMG400018235	1:79,098,011:79,100,736	1836	6	611	12	M1BJX1	NPF4.5
<i>StNRT07</i>	PGSC0003DMG400020708	1:79,794,672:79,799,593	1818	6	605	12	M1BUV8	NPF4.3
<i>StNRT08</i>	PGSC0003DMG400006913	2:26,506,508:26,508,700	1275	2	424	11	M1A9H2	NRT2.7
<i>StNRT09</i>	PGSC0003DMG402021183	2:29,139,179:29,143,017	1845	5	614	12	M1BWK6	NPF4.5
<i>StNRT10</i>	PGSC0003DMG400001393	2:46,706,587:46,710,386	1539	4	512	10	M0ZLV3	NPF5.8
<i>StNRT11</i>	PGSC0003DMG400027817	3:917,915:925,305	1794	7	597	9	M1CNW8	NPF4.5
<i>StNRT12</i>	PGSC0003DMG400018129	3:51,866,265:51,873,338	996	4	331	2	M1BJG7	NRT3.2
<i>StNRT13</i>	PGSC0003DMG400025334	3:53,073,791:53,078,316	1770	5	589	12	M1CDI3	NPF4.6
<i>StNRT14</i>	PGSC0003DMG400025339	3:53,135,911:53,138,548	1788	3	595	11	M1CDJ5	NPF2.8
<i>StNRT15</i>	PGSC0003DMG400025337	3:53,160,486:53,162,884	1758	4	585	10	M1CDJ3	NPF2.11
<i>StNRT16</i>	PGSC0003DMG400025336	3:53,169,753:53,173,630	1776	4	591	12	M1CDJ2	NPF2.11
<i>StNRT17</i>	PGSC0003DMG403002548	3:59,987,657:59,990,169	1404	1	467	8	M0ZRJ9	NPF1.2
<i>StNRT18</i>	PGSC0003DMG400002865	4:388,737:394,987	1755	5	584	12	M0ZT00	NPF4.6
<i>StNRT19</i>	PGSC0003DMG400011693	4:68,832,667:68,834,950	1758	5	585	12	M1AUE0	NPF4.5
<i>StNRT20</i>	PGSC0003DMG400011692	4:68,837,821:68,840,580	1749	5	582	11	M1AUD9	NPF4.5
<i>StNRT21</i>	PGSC0003DMG400011691	4:68,842,006:68,845,978	1056	2	351	4	M1AUD6	NPF4.5
<i>StNRT22</i>	PGSC0003DMG400014539	5:2,690,484:2,695,466	1743	4	580	12	M1B5Q1	NPF3.1
<i>StNRT23</i>	PGSC0003DMG400017621	5:6,000,174:6,006,202	1728	5	575	12	M1BHM6	NPF4.6
<i>StNRT24</i>	PGSC0003DMG400017620	5:6,018,942:6,027,657	1884	5	627	12	M1BHM4	NPF4.6
<i>StNRT25</i>	PGSC0003DMG400017637	5:6,118,146:6,121,903	1821	4	606	12	M1BHP7	NPF2.13
<i>StNRT26</i>	PGSC0003DMG400015595	5:7,359,482:7,362,610	1758	4	585	10	M1B9N3	NPF1.2
<i>StNRT27</i>	PGSC0003DMG400015572	5:7,368,720:7,371,036	1761	4	586	11	M1B9K1	NPF1.2
<i>StNRT28</i>	PGSC0003DMG400015594	5:7,392,962:7,395,545	1758	4	585	11	M1B9N2	NPF1.2
<i>StNRT29</i>	PGSC0003DMG400015593	5:7,410,649:7,413,350	1740	4	579	9	M1B9N0	NPF1.2
<i>StNRT30</i>	PGSC0003DMG400015592	5:7,415,726:7,420,927	1200	5	399	4	M1B9M9	NPF1.2
<i>StNRT31</i>	PGSC0003DMG400015591	5:7,434,852:7,436,872	1791	4	596	10	M1B9M7	NPF1.2
<i>StNRT32</i>	PGSC0003DMG400004134	5:7,524,932:7,526,789	1659	3	552	9	M0ZYA8	NPF1.2
<i>StNRT33</i>	PGSC0003DMG400004133	5:7,528,121:7,529,797	1317	4	438	9	M0ZYA7	NPF1.2
<i>StNRT34</i>	PGSC0003DMG400016996	5:9,714,775:9,717,165	1503	2	500	11	M1BF74	NRT2.5
<i>StNRT35</i>	PGSC0003DMG400025395	6:8,311,895:8,316,339	1836	4	611	9	M1CDS5	NPF3.1
<i>StNRT36</i>	PGSC0003DMG400015734	6:11,860,378:11,862,454	1575	4	524	10	M1BA89	NRT2.4
<i>StNRT37</i>	PGSC0003DMG400014054	6:31,438,515:31,443,669	1746	4	581	10	M1B3Q0	NPF5.1
<i>StNRT38</i>	PGSC0003DMG400013815	6:43,655,569:43,659,228	1779	5	592	12	M1B2Y3	NPF6.3
<i>StNRT39</i>	PGSC0003DMG401031322	6:47,048,581:47,054,364	1707	4	568	11	M1D3I7	NPF5.10
<i>StNRT40</i>	PGSC0003DMG400027083	6:52,407,554:52,410,676	1737	5	578	8	M1CKV3	NPF2.11
<i>StNRT41</i>	PGSC0003DMG402011998	6:55,526,924:55,528,285	1062	3	353	8	M1AVH0	NRT2.4
<i>StNRT42</i>	PGSC0003DMG401011998	6:55,535,257:55,537,119	1593	3	530	11	M1AVG9	NRT2.4
<i>StNRT43</i>	PGSC0003DMG400007508	6:55,971,792:55,976,592	1521	3	506	11	M1AC21	NPF2.3
<i>StNRT44</i>	PGSC0003DMG400007509	6:55,984,006:55,986,374	1641	4	546	12	M1AC22	NPF2.7
<i>StNRT45</i>	PGSC0003DMG400007510	6:55,995,187:55,997,140	1011	2	336	6	M1AC23	NPF2.7
<i>StNRT46</i>	PGSC0003DMG400007511	6:56,003,194:56,006,176	1662	4	553	10	M1AC24	NPF2.7
<i>StNRT47</i>	PGSC0003DMG400007512	6:56,007,197:56,008,144	630	1	209	5	M1AC25	NPF2.7
<i>StNRT48</i>	PGSC0003DMG400006606	6:56,077,004:56,080,118	1659	4	552	11	M1A8A5	NPF2.7
<i>StNRT49</i>	PGSC0003DMG400030438	6:57,021,923:57,025,562	1725	5	574	10	M1CZN4	NPF8.1
<i>StNRT50</i>	PGSC0003DMG400022993	7:5,773,748:5,778,627	1755	5	584	10	M1C3Y0	NPF8.1
<i>StNRT51</i>	PGSC0003DMG400029242	7:17,735,905:17,738,710	1743	6	580	12	M1CUQ8	NPF6.2
<i>StNRT52</i>	PGSC0003DMG402000668	7:45,033,113:45,036,254	1782	5	593	11	M0ZJ30	NPF6.4
<i>StNRT53</i>	PGSC0003DMG400012479	8:479,008:483,551	1773	5	590	11	M1AXI0	NPF6.3

Table 1 (continued)

Gene No	Reference gene ID	Chr location	Cds length/bps	exons	Protein length/aa	TMHs	UniProt	Protein Name
StNRT54	PGSC0003DMG400020514	8:1,178,361:1,184,079	1797	5	598	10	M1BU28	NPF7.3
StNRT55	PGSC0003DMG400029084	8:39,883,276:39,886,234	1593	3	530	9	M1CU67	NPF1.2
StNRT56	PGSC0003DMG400004795	8:52,567,376:52,571,923	1773	5	590	11	M1A0X7	NPF6.3
StNRT57	PGSC0003DMG400042635	9:395,745:398,087	1713	4	570	10	M1DR64	NPF8.1
StNRT58	PGSC0003DMG400020318	9:402,706:405,477	1701	4	566	10	M1BT78	NPF8.1
StNRT59	PGSC0003DMG400024268	9:1,176,786:1,180,857	1773	4	590	12	M1C8Y6	NPF5.6
StNRT60	PGSC0003DMG400023080	9:8,313,196:8,315,592	1689	4	562	10	M1C478	NPF8.2
StNRT61	PGSC0003DMG400003820	9:50,738,225:50,742,637	1797	4	598	12	M0ZWW9	NPF5.2
StNRT62	PGSC0003DMG400017204	9:56,806,392:56,815,251	1827	4	608	10	M1BFZ3	NPF5.2
StNRT63	PGSC0003DMG400004329	10:18,706,038:18,709,729	1752	6	583	12	M0ZZ02	NPF4.5
StNRT64	PGSC0003DMG400024120	10:37,692,715:37,696,235	1794	5	597	10	M1C8B9	NPF7.3
StNRT65	PGSC0003DMG400011085	10:54,161,108:54,166,759	1770	4	589	11	M1ARQ3	NPF5.6
StNRT66	PGSC0003DMG400011083	10:54,190,067:54,192,530	1794	4	597	10	M1ARQ0	NPF5.6
StNRT67	PGSC0003DMG400042160	10:55,059,494:55,061,848	1605	3	534	10	M1DQ45	NRT2.4
StNRT68	PGSC0003DMG400028026	11:19,452,106:19,456,164	1962	3	653	12	M1CPM1	NPF6.1
StNRT69	PGSC0003DMG400032801	11:28,624,806:28,628,332	1359	3	452	9	M1D677	NPF4.4
StNRT70	PGSC0003DMG400001145	11:41,898,143:41,899,905	1593	3	530	11	M0ZKW4	NRT2.4
StNRT71	PGSC0003DMG400019674	11:41,938,022:41,939,905	1578	3	525	11	M1BQM6	NRT2.4
StNRT72	PGSC0003DMG400019675	11:41,946,330:41,947,803	1305	3	434	10	M1BQM7	NRT2.4
StNRT73	PGSC0003DMG400025387	11:44,387,763:44,389,854	1581	3	526	12	M1CDQ9	NPF2.7
StNRT74	PGSC0003DMG400025389	11:44,400,838:44,404,224	1668	4	555	12	M1CDR2	NPF2.7
StNRT75	PGSC0003DMG400000303	12:4,387,465:4,391,552	1701	4	566	9	M0ZHD1	NPF1.2
StNRT76	PGSC0003DMG400004291	12:6,329,642:6,332,333	1779	5	592	11	M0ZYU5	NPF6.4
StNRT77	PGSC0003DMG400014449	12:48,920,928:48,925,553	1758	5	585	10	M1B5E2	NPF8.3
StNRT78	PGSC0003DMG400014473	12:48,944,132:48,959,296	1662	4	553	10	M1B5H8	NPF8.3
StNRT79	PGSC0003DMG400026455	12:50,643,015:50,647,292	1800	4	599	10	M1CIA7	NPF1.2
StNRT80	PGSC0003DMG400039085	12:56,962,543:56,967,199	1860	5	619	11	M1DIB6	NPF8.2
StNRT81	PGSC0003DMG400029396	12:57,637,416:57,642,347	1731	4	576	10	M1CVF8	NPF1.2

The reference gene ID is the of Ensembl genome gene ID. UniProt is the protein number corresponding to the longest transcript (Table S1). Cds, coding sequence. TMHs, transmembrane helices

the major conserved motifs were Motif 19 and Motif 20. Notably, StNRT67, StNRT70 and StNRT42 also contained Motif 15. Except for StNRT08, the remaining eight high-affinity StNRT proteins in C2 clade contained two Motifs 20 repeats. This distinct pattern of conserved motifs and structural domains in C2 clade differentiates it from the NRTs in the other three clades.

Gene structure and cis- element analysis of StNRT genes

The structure of the *StNRT* genes, including the number and arrangement of introns and exons, was summarized in Table S1 and depicted in Fig. 3B. The length of *StNRT* gene sequences ranged from 845 to 3894 base pairs (bps), with coding regions spanning 627 to 1962 bps. The number of exons varied significantly among the *StNRT* genes, ranging from 1 to 7. Specifically, *StNRT02*, *StNRT17*, and *StNRT47* were characterized by a single exon, while 33 *StNRT* genes contained 4 exons and 22 *StNRT* genes had 5 exons.

We identified 55 *cis*-elements within the 1000 bps region upstream of the transcription start site of the *StNRT* gene (Fig. 3A, Table 2, and Table S5). Among

these, 35 *cis*-elements were associated with environmental responses, including light-responsive elements, MeJA-responsive elements, and defense and adversity-responsive elements. Light-responsive elements were the most abundant and diverse, comprising Box4, G-box, GT1-motif, TCT-motif, and GATA-motif. Additionally, nine *cis*-elements were linked to phytohormone responses, including elements for abscisic acid responsiveness, gibberellin responsiveness, salicylic acid responsiveness, zein metabolism regulation, and auxin responsiveness. We also identified CAT-box elements for meristem expression, circadian elements for circadian control, and MBS elements for MYB binding sites involved in drought induction.

In the promoter regions of 81 *StNRT* genes, a total of 931 *cis*-elements were identified, predominantly associated with light responsiveness, hormone signaling, and stress adaptation. High-frequency light-responsive elements, such as Box 4 and G-box, were found in 58 and 51 *StNRT* genes, suggesting a potential photoregulation of nitrate transport to synchronize with photosynthesis. Furthermore, the coexistence of ABRE (ABA-responsive)

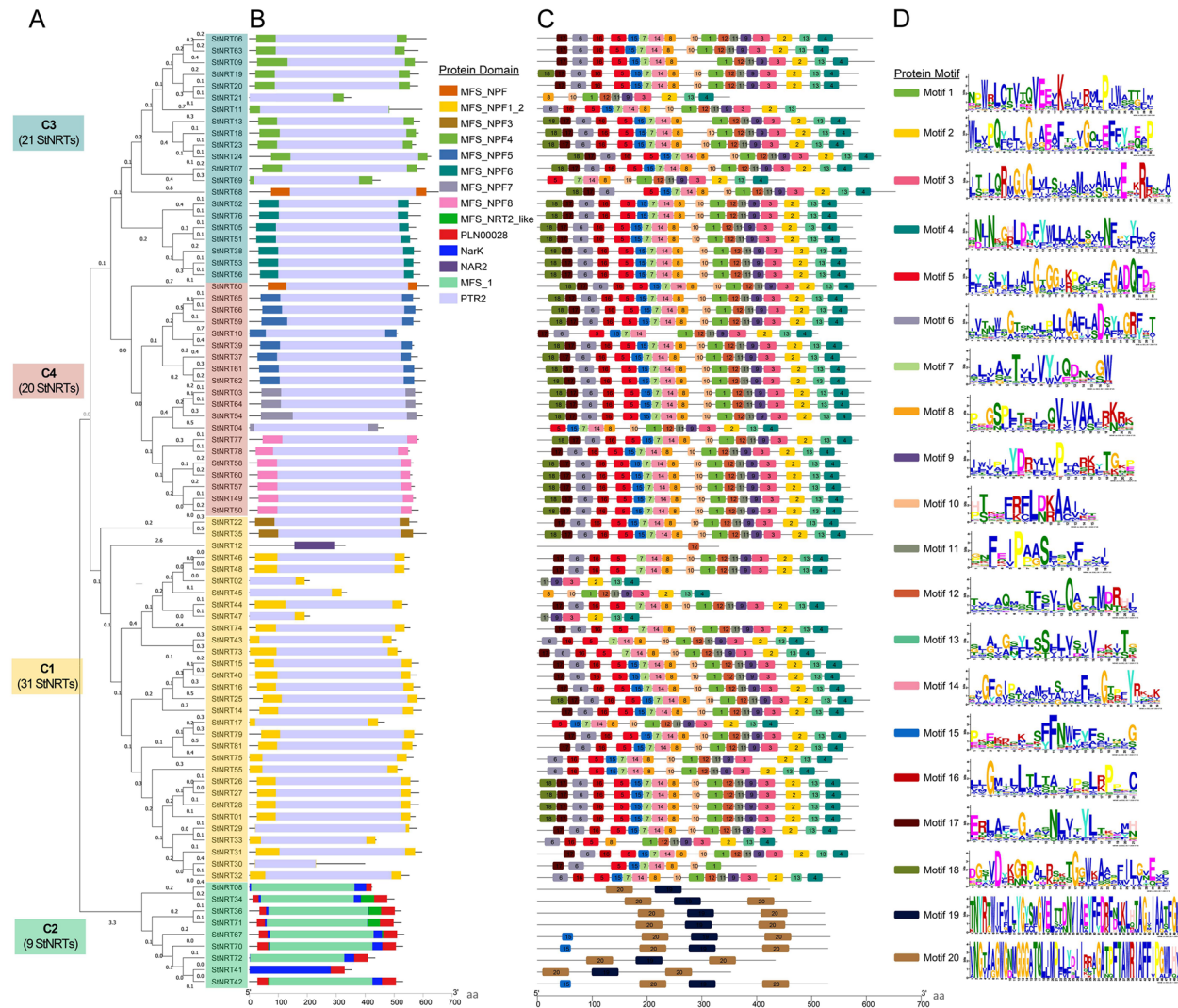


Fig. 2 Phylogenetic tree, conserved domains and motifs of StNRT proteins. **A** Evolutionary tree of StNRT proteins. Sequences were aligned using MUSCLE, and the phylogenetic tree was constructed in MEGA11 with the Maximum likelihood (ML) method. The bootstrap value was 1000 replications. StNRT proteins clustered into four clades (C1 ~ C4), indicated by colored backgrounds. **B** Conserved domains of StNRT proteins, identified via NCBI Batch CD-Search. **C** Distribution of 20 conserved motifs predicted by MEME Suite v5.5.7. **D** Sequence logos of the 20 motifs. Motif 19 and Motif 20 were only present in high affinity NRT2 proteins. Phylogenetic tree, domains and motifs were visualized using TBtools v2.1.36

and GARE-motif (gibberellin-responsive) indicates a balance between stress response and growth through NRT modulation. The Enrichment of ARE (anaerobic induction) and TC-rich repeats (defense response) suggests that *StNRTs* may play a role in nitrogen redistribution under conditions of hypoxia or pathogen attack. Additionally, the presence of LTR (low-temperature response elements) in certain *StNRTs* implies their potential function in maintaining nitrate transport during cold stress. Among the identified cis-elements, those related to light responsiveness, abscisic acid responsiveness, anaerobic induction, and MeJA-responsiveness were the most prevalent, highlighting the critical regulation of *StNRT* genes by environmental signals such as light, ABA, oxygen deficiency, and secondary metabolites.

Phylogenetic and synteny analysis of *NRT* genes

Analyses of intraspecific collinearity and evolutionary divergence in potato (Fig. 4A, Table S6) identified 14 collinear gene pairs containing *StNRT* genes, among which Ka/Ks ratios were calculated for 10 pairs (Table 3). The analysis further revealed eight segmental duplication clusters (SDG1–SDG8) and two tandem duplication clusters (TDG1–TDG2) involving *StNRT* genes. Among these, eight pairs were segmentally duplicated, while *StNRT53/StNRT56* and *StNRT61/StNRT62* were predicted to be tandemly duplicated. Each cluster comprises two duplicated genes, and their evolutionary divergence was quantified using non-synonymous (Ka) and synonymous (Ks) substitution rates, as well as Ka/Ks ratios. All clusters exhibited Ka/Ks ratios < 1, indicating that the

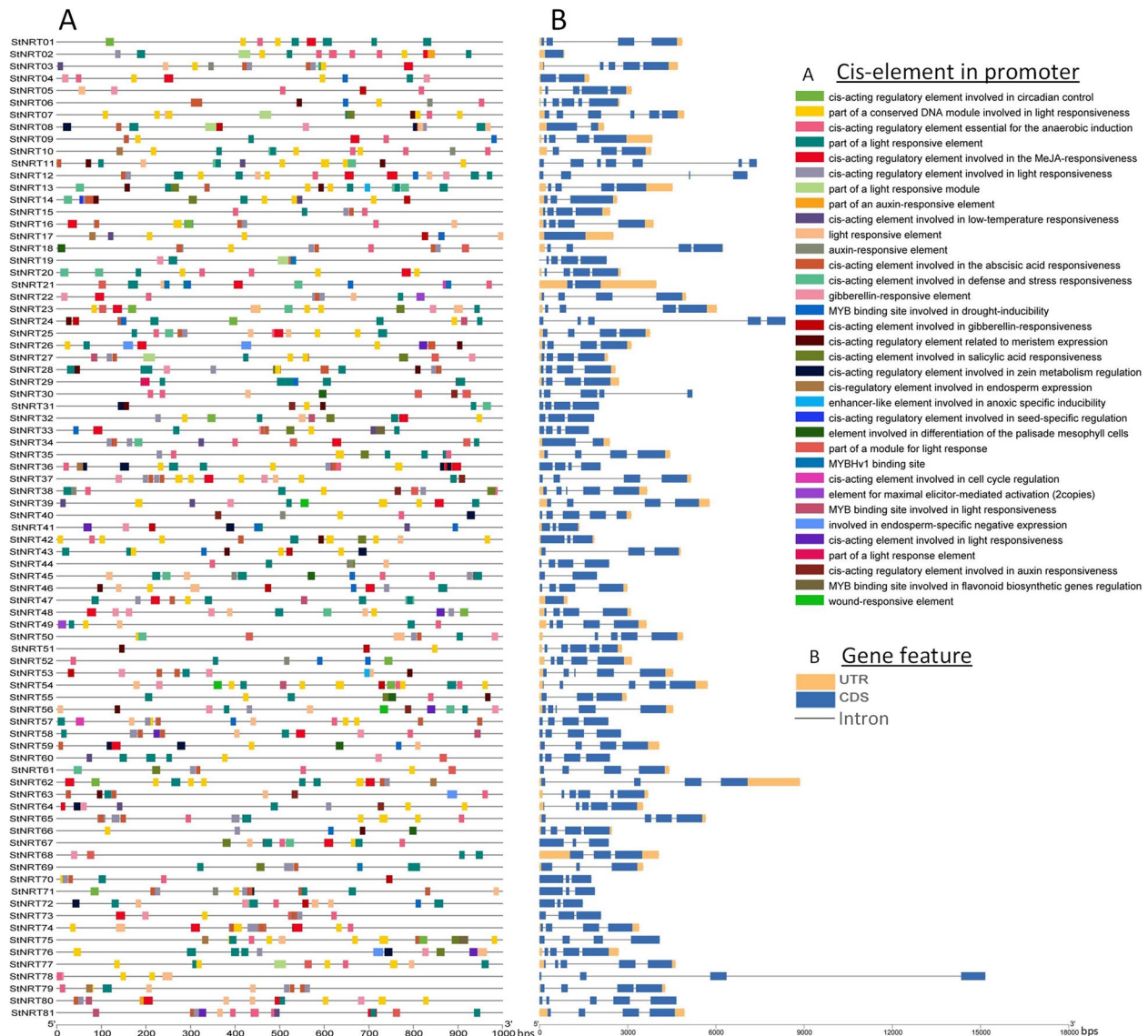


Fig. 3 Gene structure and *cis*-regulatory elements of *StNRT* genes. **A.** Predicted *Cis*-elements within the 1 kb promoter regions of *StNRT* genes, identified using PlantCARE. **B.** Exon—intron distribution of *StNRT* genes. Gene structures and *cis*-elements distributions were visualized using TBtools v2.136

uplicated *StNRT* genes have undergone purifying selection to maintain functional integrity. Notably, SDG1 (*StNRT41/StNRT67*) displayed the lowest K_a/K_s ratio (0.0746), suggesting strong purifying selection. The high K_s value (2.5535) in this cluster implies that the duplication event occurred relatively early in evolutionary history. In contrast, SDG5 (*StNRT17/StNRT79*) exhibited the highest K_a/K_s ratio (0.2499) among segmental duplications, indicating slightly relaxed purifying selection compared to other clusters. The K_a/K_s ratios < 1 across all clusters underscore that non-synonymous substitutions are generally deleterious and are not fixed, reflecting strong selective pressure to retain the original functions of these genes. This conservation is likely attributable to the critical roles of *StNRT* genes in nitrate transport

and stress adaptation. The K_s values provide insights into the timing of duplication events. Clusters with higher K_s values (e.g., SDG1, TDG2) likely represent older duplications, while those with lower K_s values (e.g., SDG7, TDG1) correspond to more recent events.

The coexistence of segmental and tandem duplications highlights the significance of gene duplication in expanding the *StNRT* gene family, potentially enhancing the adaptability of potato to diverse environmental conditions. These findings collectively emphasize the evolutionary constraints and functional conservation of *StNRT* genes in potato.

To investigate the evolutionary conservation of *NRT* genes across species, we performed inter-species synteny analysis by comparing potato with tomato, *Arabidopsis*,

Table 2 List of *cis*-elements identified in the promoter regions of *StNRT* genes

Cis-element name	Sequence	Function	Count	Genes
Box 4	ATTAAT	part of a conserved DNA module involved in light responsiveness	126	58
G-box	TACGTG	cis-acting regulatory element involved in light responsiveness	99	51
ABRE	ACGTG	cis-acting element involved in the abscisic acid responsiveness	85	44
ARE	AAACCA	cis-acting regulatory element essential for the anaerobic induction	67	44
GT1-motif	GGTTAAT	light responsive element	47	33
TCT-motif	TCTTAC	part of a light responsive element	42	37
CGTCA-motif	CGTCA	cis-acting regulatory element involved in the MeJA-responsiveness	40	33
TGACG-motif	TGACG	cis-acting regulatory element involved in the MeJA-responsiveness	40	33
GATA-motif	AAGATAAGATT	part of a light responsive element	28	23
TC-rich repeats	GTTTCTTAC	cis-acting element involved in defense and stress responsiveness	23	19
CAT-box	GCCACT	cis-acting regulatory element related to meristem expression	22	18
MBS	CAACTG	MYB binding site involved in drought-inducibility	22	18
P-box	CCTTTTG	gibberellin-responsive element	20	18
TCA-element	CCATCTTTT	cis-acting element involved in salicylic acid responsiveness	19	19
AE-box	AGAAACAA	part of a module for light response	18	16
O2-site	GATGATGTGG	cis-acting regulatory element involved in zein metabolism regulation	16	11
TATC-box	TATCCCA	cis-acting element involved in gibberellin-responsiveness	16	16
GA-motif	ATAGATAA	part of a light responsive element	15	15
GARE-motif	TCTGTTG	gibberellin-responsive element	14	13
LTR	CCGAAA	cis-acting element involved in low-temperature responsiveness	14	12
circadian	CAAAGATATC	cis-acting regulatory element involved in circadian control	11	11
I-box	AGATAAGG	part of a light responsive element	11	10
chs-CMA1a	TTACTTAA	part of a light responsive element	9	8
TGA-element	AACGAC	auxin-responsive element	9	9
ACE	CTAACGTATT	cis-acting element involved in light responsiveness	8	8
AuxRR-core	GGTCCAT	cis-acting regulatory element involved in auxin responsiveness	8	8
MRE	AACCTAA	MYB binding site involved in light responsiveness	8	8
TCCC-motif	TCTCCCT	part of a light responsive element	8	8
ATCT-motif	AATCTAATCC	part of a conserved DNA module involved in light responsiveness	7	7
GCN4_motif	TGAGTCA	cis-regulatory element involved in endosperm expression	7	7
3-AF1 binding site	TAAGAGAGGAA	light responsive element	6	5
AT1-motif	AATTATTTTTTATT	part of a light responsive module	6	6
HD-Zip 1	CAAT(A/T) ATTG	element involved in differentiation of the palisade mesophyll cells	6	6
Sp1	GGGCGG	light responsive element	5	4
AACA_motif	TAACAACTCCA	involved in endosperm-specific negative expression	4	3
CCAAT-box	CAACGG	MYBHv1 binding site	4	3
chs-CMA2a	TCACTTGA	part of a light responsive element	4	4
LAMP-element	CTTTATCA	part of a light responsive element	4	4
ATC-motif	AGTAATCT	part of a conserved DNA module involved in light responsiveness	3	3
Box II	TGGTAATAA	part of a light responsive element	3	3
Gap-box	CAAATGAA(A/G) A	part of a light responsive element	3	3
TGA-box	TGACGTAA	part of an auxin-responsive element	3	3
WUN-motif	AAATTCCT	wound-responsive element	3	3
AT-rich sequence	TAAATACT	element for maximal elicitor-mediated activation (2copies)	2	2
GC-motif	CCCCCG	enhancer-like element involved in anoxic specific inducibility	2	2
GTGGC-motif	GATTCTGTGGC	part of a light responsive element	2	2
L-box	ATCCCACCTAC	part of a light responsive element	2	2
MBSI	aaaAaaC(G/C) GTTA	MYB binding site involved in flavonoid biosynthetic genes regulation	2	2
MSA-like	TCCAACGGT	cis-acting element involved in cell cycle regulation	2	2
4 cl-CMA1b	ATTCCGATAAACT	light responsive element	1	1
CAG-motif	GAAAGGCAGAC	part of a light response element	1	1
chs-Unit 1 m1	ACCTAACCCGC	part of a light responsive element	1	1
Pc-CMA2c	GCCCCACGCA	part of a light responsive element	1	1

Table 2 (continued)

Cis-element name	Sequence	Function	Count	Genes
RY-element	CATGCATG	cis-acting regulatory element involved in seed-specific regulation	1	1
sbp-CMA1c	CTTTATCTCTCCA	part of a light responsive element	1	1
All			931	81

The “sequence” columu denotes the specific cis-elements identified in the promoter sequences, the “Count” column indicates the total number of occurrences of each cis-element across all promoter sequences, and the “Genes” column specifies the number of StNRT genes in which a particular cis-element was identified

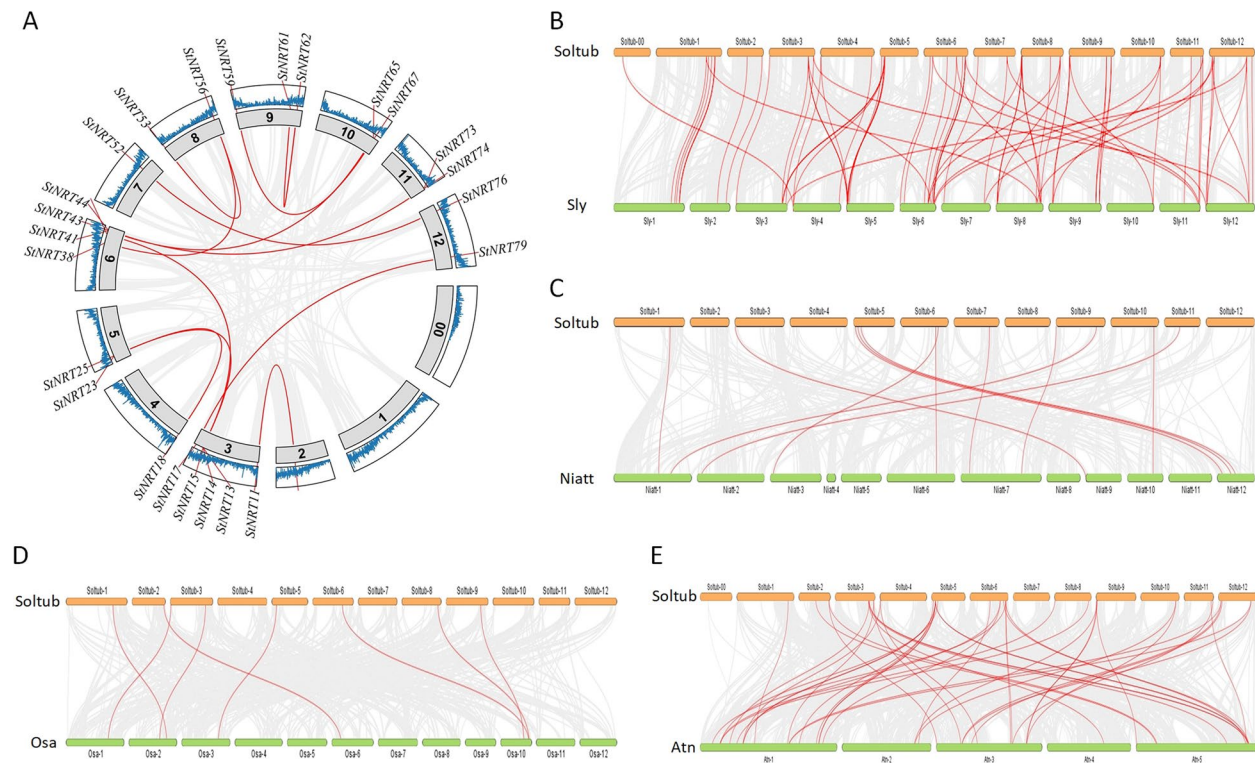


Fig. 4 Intrasppecific and interspecific synteny analysis. **A** Collinearity analysis within potato genome. Red lines represent syntenic gene pairs between SDGs (Segmental Duplicate Genes) and TDGs (Tandem Duplicate Genes). **B** Synteny between *S. tuberosum* (Soltub) and *S. lycopersicum* (Sly). **C** Synteny between *S. tuberosum* and *Nicotiana attenuata* (Niatt). **D** Synteny between *S. tuberosum* and *Oryza sativa* (Osa, rice). **E** Synteny between *S. tuberosum* and *Arabidopsis thaliana* (Atn). In panels B~E, red line represents syntenic relationships of *StNRT* genes with the respective species genomes (Sly, 72 pairs; Niatt, 12 pairs; Osa, 8 pairs; Atn, 45 pairs; see Table S6). Gray lines represent syntenic relationships of other genes between the genomes

tobacco, and rice (Fig. 4B~E). The results demonstrated that the number of collinear *NRT* gene pairs was significantly higher between potato and tomato (72 pairs) than between potato and *Arabidopsis* (45 pairs), tobacco (12 pairs), or rice (8 pairs) (Table S6). These findings highlight stronger synteny and evolutionary conservation between potato and tomato, likely due to their close phylogenetic relationship. In contrast, the reduced collinearity with more distantly related species such as *Arabidopsis*, tobacco, and rice suggests divergent evolutionary trajectories in the *NRT* gene family.

Synteny analysis demonstrated strong evolutionary conservation between potato *StNRT* genes and tomato *SINRT* genes. To further elucidate phylogenetic relationships, we performed comparative analysis of protein sequences from 81 *StNRT*s (potato), 85 *SINRT*s (tomato), and 38 *AtNRT*s (*Arabidopsis thaliana*), encompassing

204 *NRT* proteins in total (Fig. 5, Table S7). A maximum likelihood phylogenetic tree was constructed, revealing four distinct clades (Clade I~Clade IV), consistent with the classification patterns observed in the *StNRT* specific phylogenetic analysis. Clade I comprised 78 *NRT*s, including canonical transporters such as *NRT1.1*, *NPF1.2*, *NPF2.7*, and *NPF3.1*. Clade II contained 23 high-affinity nitrate transporters, predominantly represented by *NRT2.4*. Clade III included 51 *NRT*s associated with the *NPF4* and *NPF6* subfamilies, while Clade IV encompassed 52 *NRT*s classified into *NPF5*, *NPF7*, and *NPF8* subfamilies. This phylogenetic structure not only reflects functional diversification within the *NRT* family but also underscores conserved evolutionary trajectories among the examined species, particularly between potato and tomato.

Table 3 List of StNRT gene tandem/segmentally duplicated gene clusters

Cluster name	Gene 1	Gene 2	Ka	Ks	Ka/Ks
SDG1	StNRT41	StNRT67	0.1905	2.5535	0.0746
SDG2	StNRT59	StNRT65	0.1281	0.7445	0.1720
SDG3	StNRT43	StNRT73	0.2960	1.3505	0.2192
SDG4	StNRT44	StNRT74	0.3036	1.2864	0.2360
SDG5	StNRT17	StNRT79	0.2093	0.8377	0.2499
SDG6	StNRT52	StNRT76	0.1112	0.8302	0.1340
SDG7	StNRT23	StNRT18	0.1104	0.8680	0.1272
SDG8	StNRT56	StNRT38	0.1450	1.1404	0.1271
TDG1	StNRT56	StNRT53	0.0934	1.0688	0.0874
TDG2	StNRT62	StNRT61	0.2595	2.2803	0.1138

The table provides a list of tandemly duplicated gene clusters (TDG) and segmentally duplicated gene clusters (SDG) involving *StNRT* genes, along with their respective Ka (non-synonymous substitution rate), Ks (synonymous substitution rate), and Ka/Ks ratios. TDG represents tandemly duplicated gene clusters, while SDG denotes segmentally duplicated gene clusters. Ka is defined as the rate of non-synonymous substitutions per non-synonymous site, and Ks represents the rate of synonymous substitutions per synonymous site. The Ka/Ks ratio, calculated as the ratio of Ka to Ks, serves as an indicator of the selective pressure acting on the duplicated genes

Gene ontology enrichment analysis of StNRT genes

GO enrichment analysis of the 81 *StNRT* genes identified 23 significantly enriched terms (Table 4, Table S8). These terms were categorized into three major GO domains. Cellular Component: Enriched terms included membrane, integral component of membrane, and intrinsic component of membrane. Biological Process: Key processes encompassed transmembrane transport, nitrogen compound transport, cellular process, and peptide transport. Molecular Function: Enriched functions featured transmembrane transporter activity, transporter activity, and nitrate transmembrane transporter activity. These results underscore the central role of *StNRT* genes in nitrate transport and related physiological processes, with additional involvement in peptide transport mechanisms, suggesting functional diversification within the gene family. Nine high-affinity nitrate transporters were enriched into the ko00910 Nitrogen metabolism Pathway.

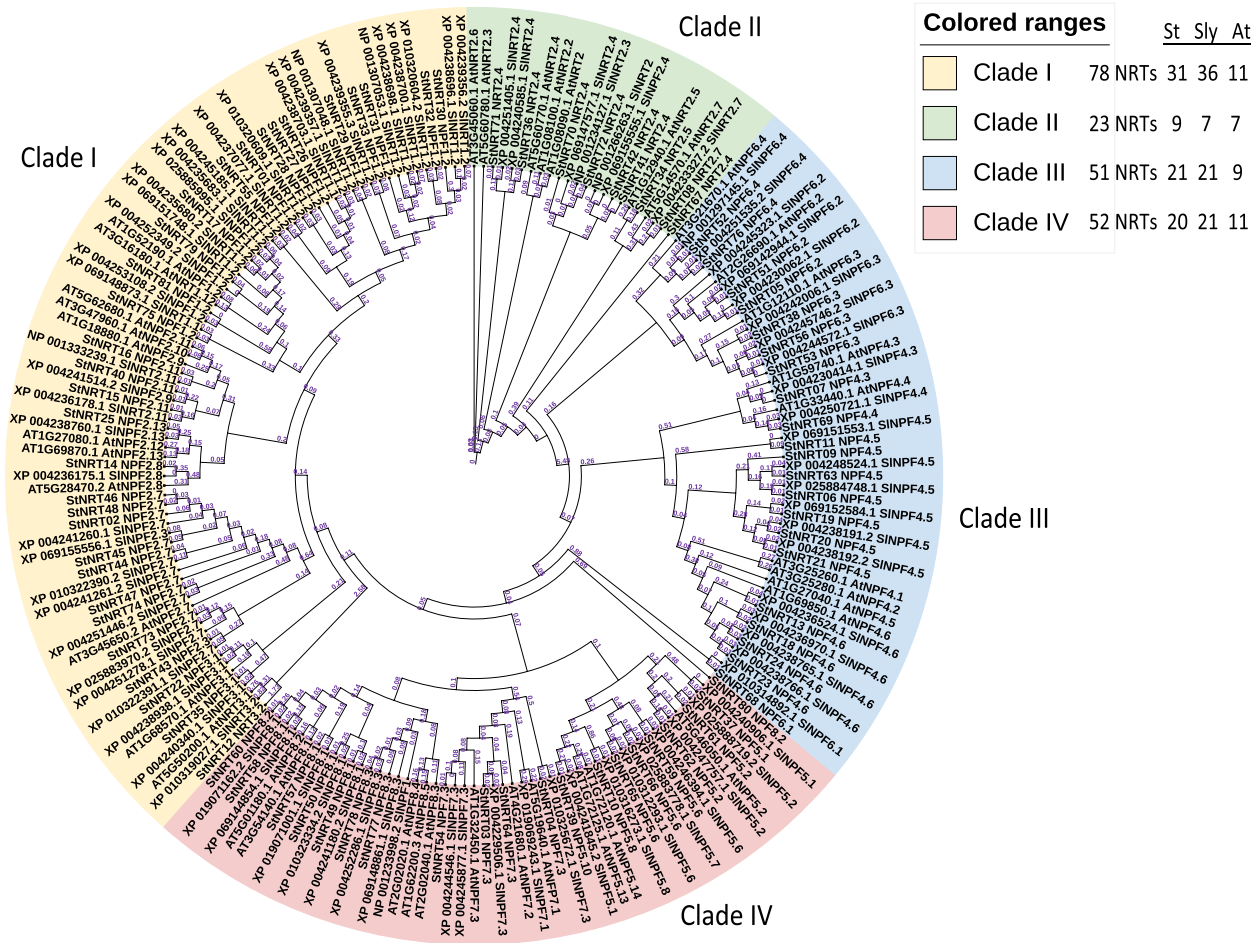


Fig. 5 Phylogenetic relationships of *NRT* genes in *Solanum tuberosum* (St; potato; StNRT30/NPF1.2), *S. lycopersicum* (Sly; tomato; XP 004239356.2/SINRT1.2), and *Arabidopsis thaliana* (At; thale cress; AT1G52190.1/AtNRT1.2). Protein sequences were aligned using MUSCLE, and a maximum likelihood (ML) tree was constructed in MEGA11 with 1000 bootstrap replicates. The 204 NRTs were classified into four clades: Clade I (78), Clade II (23), Clade III (51), and Clade IV (52). Branch numbers represent evolutionary distances. The final tree was visualized and annotated in iTOL v7.1

Table 4 Gene Ontology analysis of StNRT

Category	GO.ID	Term	Gene	Total	P value	FDR	Representative Genes
MF	GO:0022857	transmembrane transporter activity	72	910	9.98E-94	7.99E-93	StNRT75, StNRT10, StNRT03
MF	GO:0005215	transporter activity	72	956	4.01E-92	1.60E-91	StNRT75, StNRT10, StNRT03
BP	GO:0055085	transmembrane transport	80	973	4.59E-89	2.52E-87	StNRT75, StNRT70, StNRT10
BP	GO:0006810	transport	80	1688	2.67E-69	7.35E-68	StNRT75, StNRT70, StNRT10
BP	GO:0051234	establishment of localization	80	1697	4.13E-69	7.57E-68	StNRT75, StNRT70, StNRT10
BP	GO:0051179	localization	80	1726	1.65E-68	2.28E-67	StNRT75, StNRT70, StNRT10
BP	GO:0006857	oligopeptide transport	25	28	2.40E-53	2.20E-52	StNRT75, StNRT56, StNRT48
BP	GO:0015833	peptide transport	25	28	2.40E-53	2.20E-52	StNRT75, StNRT56, StNRT48
BP	GO:0042886	amide transport	25	32	2.42E-50	1.90E-49	StNRT75, StNRT56, StNRT48
CC	GO:0016021	integral component of membrane	81	6219	1.82E-22	2.33E-21	StNRT75, StNRT70, StNRT10
CC	GO:0031224	intrinsic component of membrane	81	6234	2.22E-22	2.33E-21	StNRT75, StNRT70, StNRT10
BP	GO:0071705	nitrogen compound transport	28	422	5.42E-21	3.73E-20	StNRT75, StNRT56, StNRT48
CC	GO:0016020	membrane	81	6865	5.74E-19	4.02E-18	StNRT75, StNRT70, StNRT10
BP	GO:0071702	organic substance transport	25	577	2.44E-14	1.49E-13	StNRT75, StNRT56, StNRT48
BP	GO:0009987	cellular process	80	9294	2.35E-09	1.29E-08	StNRT75, StNRT70, StNRT10
BP	GO:0015706	nitrate transmembrane transport	3	4	1.16E-06	5.33E-06	StNRT08, StNRT34, StNRT12
BP	GO:1,902,025	nitrate import	3	4	1.16E-06	5.33E-06	StNRT08, StNRT34, StNRT12
BP	GO:0098661	inorganic anion transmembrane transport	3	32	1.26E-03	0.0053	StNRT08, StNRT34, StNRT12
BP	GO:0009624	response to nematode	2	10	1.94E-03	0.0076	StNRT37, StNRT07
MF	GO:0015112	nitrate transmembrane transporter activity	1	1	4.20E-03	0.0111	StNRT08
BP	GO:0098656	anion transmembrane transport	3	57	6.59E-03	0.0231	StNRT08, StNRT34, StNRT12
BP	GO:0010336	gibberellic acid homeostasis	1	1	6.72E-03	0.0231	StNRT22
BP	GO:0015698	inorganic anion transport	3	66	9.87E-03	0.0319	StNRT08, StNRT34, StNRT12

Gene refers to the StNRT gene numbers enriched in the GO entry. MF molecular function, BP biological process, CC cellular component

Tissue-specific expression profile of StNRT genes

To elucidate the spatial expression patterns of *StNRT* genes, we analyzed RNA-seq data from the SpudDB database across 21 potato tissues, including whole plant, root, stem, leaf, petiole, shoot apex, stolon, young tuber, mature tuber, tuber cortex, tuber peel, tuber pith, tuber sprout, flower, sepal, petal, stamen, carpel, immature whole fruit, mature whole fruit, inside of fruit and cal-
lus (Fig. 6A). These findings revealed distinct tissue-specific expression profiles. Most high-affinity *StNRT* genes exhibited low expression under high nitrogen conditions. Notably, *StNRT2.4* showed elevated expression in roots and tuber peel. *StNRT08* (*NRT2.7*) was strongly expressed in stems, leaves, flowers (particularly carpels), and fruits, but not expressed in roots. *StNRT52* (*NPF6.4*), *StNRT39* (*NPF5.10*), *StNRT49* (*NPF8.1*), *StNRT35* (*NPF3.1*), *StNRT22* (*NPF3.1*), and *StNRT25* (*NPF2.13*) displayed high expression at the whole-plant level. *StNRT39*, *StNRT49*, *StNRT22*, *StNRT35*, and *StNRT25* were ubiquitously expressed across all the potato tissues. The results of the specific expression analysis revealed that *StNPF1.2* was expressed in the stolons. Floral-specific expression included *StNPF8.1*, *StNPF8.2*, and *StNPF2.8* (flowers), *StNPF4.6* (sepals), and *StNPF6.4*, *StNPF7.3*, and *StNPF8.1* (stamens). Petiole-specific expression was observed for *StNPF4.5*, *StNPF5.6*, *StNPF6.3*, *StNPF2.7*, and *StNPF5.8*. Stem-enriched genes included *StNPF4.5*, *StNPF6.1*, *StNPF2.11*, *StNPF6.2*, and *StNPF6.3*. Tuber-related

expression: *StNRT2.4* was uniquely expressed in mature tubers and tuber peel (Fig. 6, Table S9).

The tissue-specific expression analysis revealed distinct spatial patterns for selected *StNRT/NPF* genes. *StNRT11* (*NPF4.5*) exhibited specific expression in fruits. *StNRT44* (*NPF2.7*) was expressed in roots and fruits. *StNRT70* (*NRT2.4*) showed specific expression in roots and tuber peel. *StNRT80* (*NPF8.2*) was expressed in flowers and tuber sprouts. In contrast, *StNRT67* (*NRT2.4*) expression was not detected in any of the examined tissues. These results highlight the functional diversification of *StNRT* genes, with specific members contributing to nitrogen transport and signaling in distinct organs, developmental stages, and environmental contexts.

Expression analysis of StNRT genes in response to nitrogen stress

Expression profiles of 81 *StNRT* genes were analyzed in roots, stems and leaves under three nitrogen levels (low, medium, and high) in the two potato cultivars (Fig. 7 and Table S10). *StNRT2.4* expression was up-regulated in roots under medium and low nitrogen conditions. Under high nitrogen conditions, *StNRT43*(*NPF2.3*) exhibited elevated expression in stems of Qingshu 9, *StNRT13* (*NPF4.6*) showed increased expression in leaves of Qingshu 9, and *StNRT75* (*NPF1.2*) displayed higher expression in stems of Xiazhai 65. Under low nitrogen conditions, *StNRT19* (*NPF4.5*) was up-regulated in

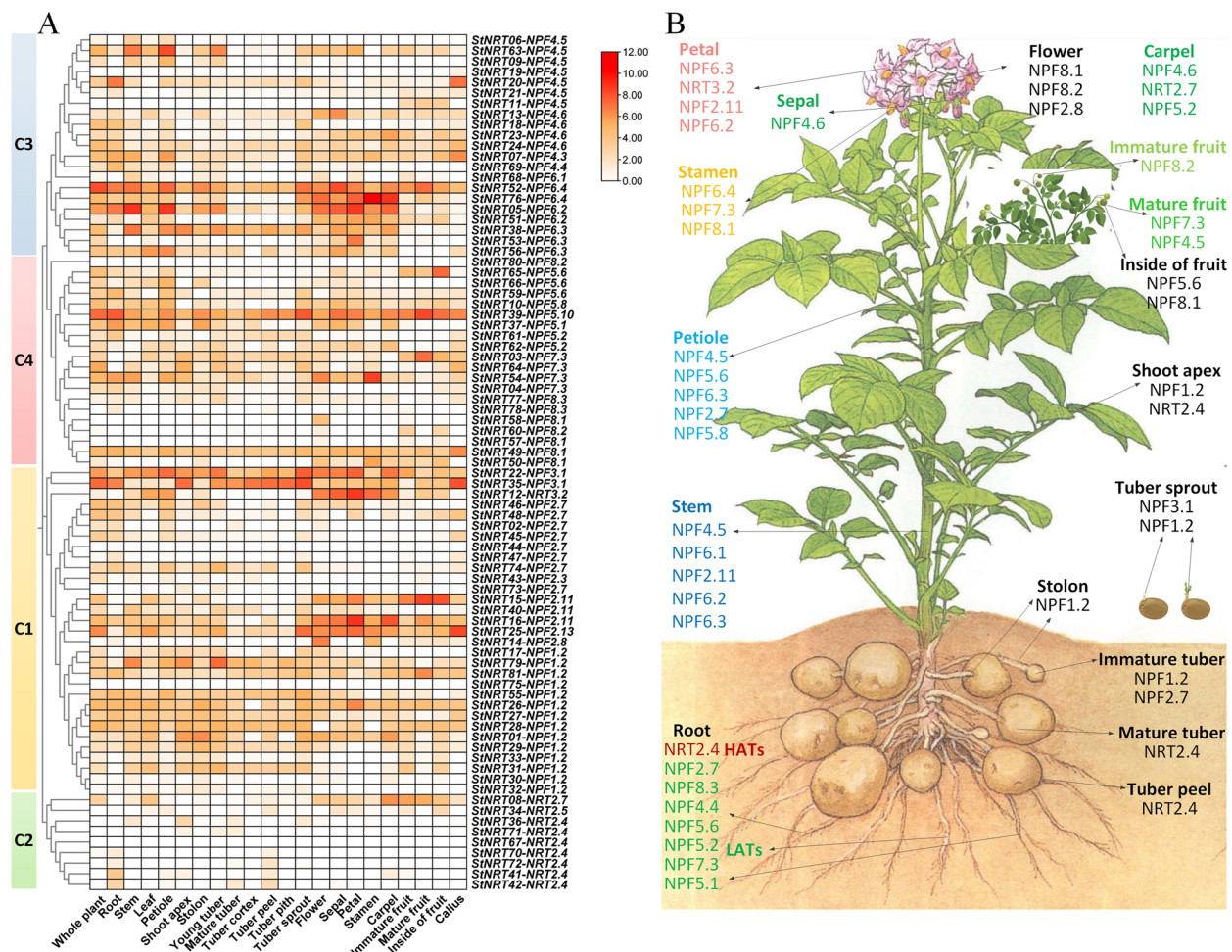


Fig. 6 Tissue-specific expression patterns of *StNRT/NPF* genes. **A** Spatial expression profile of *StNRT/NPF* genes across 21 potato tissues: whole plant, root, stem, leaf, petiole, shoot apex, stolon, young tuber, mature tuber, tuber cortex, tuber peel, tuber pith, tuber sprout, flower, sepal, petal, stamen, carpel, immature whole fruit, mature whole fruit, inside of fruit, and callus. Heatmap color intensity reflects normalized expression levels (log2-transformed). Clusters C1–C4 (left) reflect the phylogenetic groupings established in Fig. 2A. **B** Tissue-specificity of *StNRT/NPF* genes quantified using tau(τ) index (Table S9), which indicates how specific or broadly expressed a gene or transcript is, within studied tissues. The τ index ranges from 0 (broadly expressed across all tissues) to 1 (exclusively expressed in a single tissue). Genes with $\tau > 0.8$ are classified as tissue-specific, while $\tau < 0.2$ indicates ubiquitous expression

roots of Qingshu 9, and *StNRT61* (*NPF5.2*) and *StNRT62* (*NPF5.2*) were up-regulated in roots of Xiazhai 65. Under medium nitrogen conditions, *StNRT21* (*NPF4.5*) was up-regulated in roots of Xiazhai 65.

StNRT38 (*NPF6.3*), *StNRT63* (*NPF4.5*), and *StNRT06* (*NPF4.5*) are highly expressed in the stems of both cultivars, suggesting their roles as nitrogen storage or remobilization transporters in stems. *StNRT12* (*NRT3.2*) exhibits high expression in both cultivars under medium and low nitrogen conditions, indicating it may function as a key gene maintaining the stability of the nitrogen uptake system. *StNRT52* (*NPF6.4*) and *StNRT22* (*NPF3.1*) show consistently high expression levels across all three nitrogen conditions (A, B, C) and all three tissues (root, stem, leaf) in both cultivars, demonstrating their broad and constitutive expression patterns in potato. *StNRT20* (*NPF4.5*) displays higher expression in roots compared

to stems and leaves. This observation is consistent with tissue-specific expression profiles identified in earlier analyses.

The heatmap of protein quantification for *StNRT* genes in root, stem and leaf tissue under three nitrogen levels was presented in Fig. 8 (Table S11). *StNRT42* (*NRT2.4*) expression in roots increased with decreasing nitrogen level. Protein levels of *StNRT22* (*NRT3.1*) and *StNRT78* (*NRT8.3*) remained relatively stable across nitrogen conditions. *StNRT33* protein exhibited the highest expression in stems of Qingshu 9 under high-nitrogen conditions, with down-regulation observed as nitrogen level decreased. *StNRT12* showed relatively low level in leaves under high nitrogen conditions.

We analyzed the relative expression of three *StNRT* genes in root, stem, and leaf tissues of two potato varieties under varying nitrogen levels (Fig. 9), and the

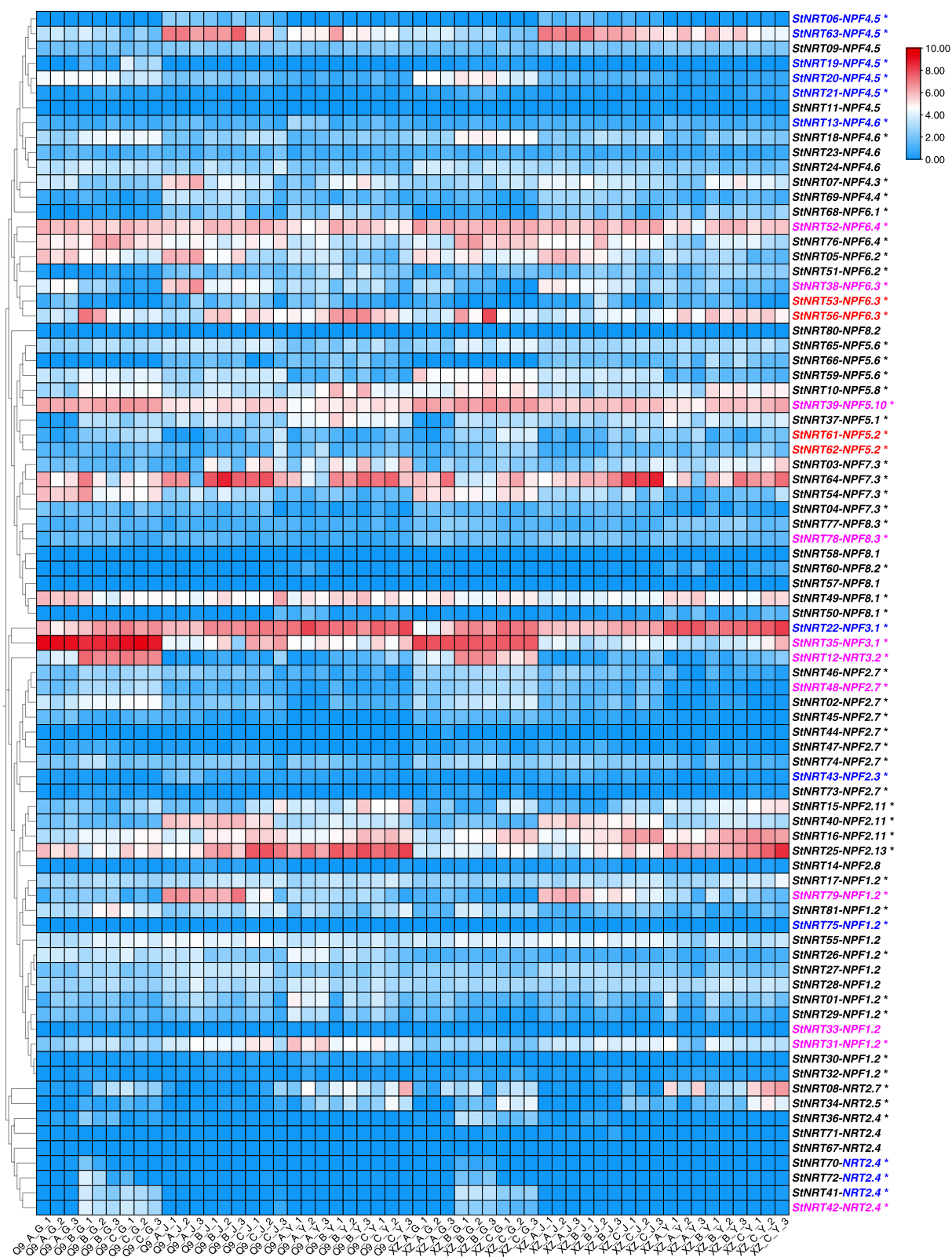


Fig. 7 (See legend on next page.)

(See figure on previous page.)

Fig. 7 Transcriptome response to nitrogen stress in potato cultivars. Heatmap with a color bar indicating normalized FPKM values of differentially expressed genes. Sample labeling: Q9, Qingshu 9; XZ, Xiazhai 65. Expression patterns of *StNRT* genes under three nitrogen levels: A, High nitrogen (40 mM NO_3^-); B, Medium nitrogen (8 mM NO_3^-); C, Low nitrogen (2 mM NO_3^-). Tissue abbreviations: G, root; J, stem; Y, leaf. Example: Q9_A_G denotes root of Qingshu 9 under high nitrogen treatment. Three biological replicates were analyzed per sample. Gene annotations by color: Red: Tandem duplicate genes (TDGs); Purple: Genes detected in the proteome; Blue: Genes discussed in the text. Significantly differential genes ($\text{FDR} < 0.05$) were marked with asterisks (*) beside gene labels

gene-specific sequences of primers used are listed in Table S12. *StNRT12* (*NAR2*) exhibited the highest expression under high nitrogen conditions and the lowest under low nitrogen, with consistent trends across all three tissues. *StNRT12* gene expression in roots and stems was down-regulated as nitrogen decreased.

StNRT70 (*NRT2.4*) showed elevated expression under low nitrogen, but reduced expression in roots under medium nitrogen. *StNRT56* (*NPF6.3*) displayed higher expression under high nitrogen, with lower expression in roots under medium nitrogen. Under medium nitrogen, both *StNRT70* and *StNRT56* exhibited reduced expression levels in roots. These highlight the differential regulation of *StNRT* genes in response to nitrogen availability, with tissue-specific expression patterns emphasizing their roles in nitrogen stress adaptation.

Discussion

StNRT gene family members in potato

In this study, we identified 81 *StNRT* genes in the potato Soltub_3.0 genome and conducted comprehensive structural and functional analyses. This represents a significant expansion compared to the 39 *StNRT* genes previously reported by Zhang et al. [23]. By comparing our results with theirs (Table S13 and Fig. S4), we found that one gene (*StNRT06* identified by Zhang, PGSC0003DMG400018193) does not belong to the NRT gene family based on its conserved domain analysis. However, the remaining *StNRT* genes reported in their study are consistent with our findings.

To further enhance the accuracy and completeness of our analysis, we also utilized the latest telomere-to-telomere (T2T) DM8 genome, which identified 96 *StNRT* genes. Notably, two genes (*StNRT01* and *StNRT02*) — previously mapped to Soltub_3.0's Chr_0 (a pseudochromosome containing unanchored scaffold sequences) — were successfully assigned to chromosomes 4 and 6 on DM8, respectively. Among the 96 *StNRT* genes identified in the DM8 genome, 9 contain the high-affinity domain PLN00028, and 2 contain the NAR2 domain (DM8C03G26330.1 and DM8C03G26340.1). These two NAR2-domain containing genes correspond to splice variants of *StNRT12*, which are known to assist high-affinity nitrate transporters in nitrate absorption and transport.

Although the DM8 genome represents the most advanced T2T assembly and has identified additional

tandem repeat genes on chromosomes 3, 5, 6, 11, and 12 (Fig. S1), its limited annotation in public databases poses challenges for downstream analysis. In contrast, the Soltub_3.0 genome, despite being less complete, is widely annotated and compatible with multiple databases (e.g., UniProt, and Smart), making it more suitable for comprehensive functional and comparative analyses. Therefore, this study primarily used the Soltub_3.0 genome as the reference genome, with the DM8 genome serving as additional files (Fig. S3, S5 ~ 10).

The 15 newly identified genes in the DM8 genome did not significantly alter phylogenetic or functional classification, as they were all assigned to existing NRT subfamilies. In the phylogenetic tree (Fig. S10), these genes are distributed across established subfamily branches without substantially altering the overall topology of major subfamilies. Our functional predictions and classification criteria for key members also apply to these new genes. Thus, while the additional NRT genes in DM8.1 enrich the family's members, they do not fundamentally change our conclusions—based on SolTub_3.0 — regarding the primary phylogenetic clustering patterns and core functional classifications of the NRT family.

This dual-genome approach not only enhances the robustness of our findings but also provides a foundation for future studies aimed at further elucidating the functional roles and regulatory mechanisms of the *StNRT* gene family in potato.

Presence of protein domains in StNRT genes

The conserved domains and motifs of the 81 *StNRT* proteins were analyzed and phylogenically divided into four clades. *StNRT* gene family expansion was driven by tandem and segmental duplication events. High-affinity NRT proteins, characterized by the conserved domains MFS_1 (Pfam07690) and NAR2 (Pfam16974), were predominantly grouped within C2 clade. Low-affinity NPF proteins, containing domains such as PTR2 (Pfam00854) and NPF (cd17351), were distributed across C1, C3, and C4 clades, with specific subfamilies (e.g., NPF6, NPF4, NPF3, NPF1_2, NPF5, NPF8, and NPF7) occupying distinct clades [22]. *StNRT12*, which contains the NAR2 domain and two transmembrane domains, plays a critical role in high-affinity nitrate transport mediated by NRT2 proteins in plants [44, 45].

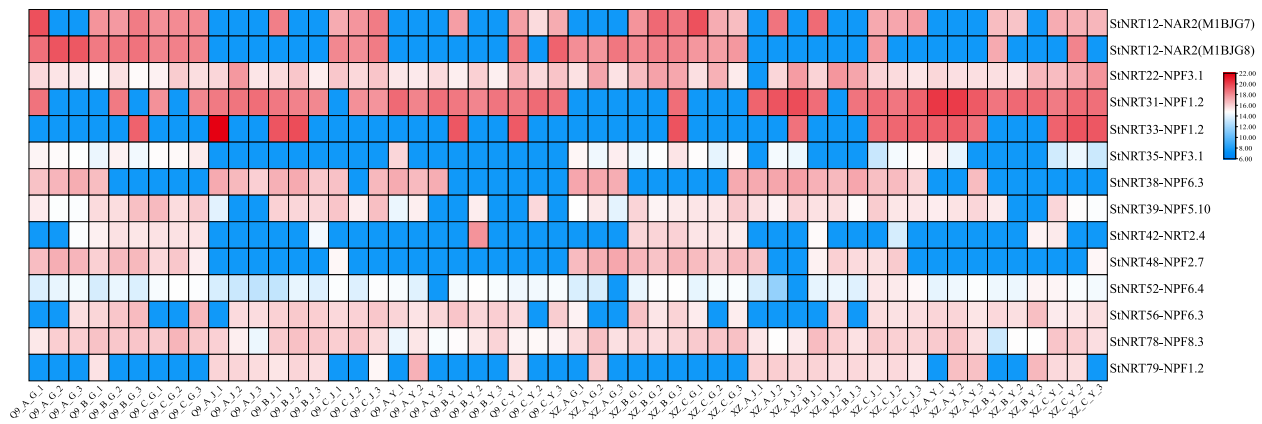


Fig. 8 Heatmap of protein quantification under nitrogen stress. The caption same as Fig. 7

Gene regulation of StNRT

The identification of 55 *cis*-elements in the promoter regions of *StNRT* genes highlights the complex regulatory network governing their expression. The predominance of light-responsive elements (e.g., Box 4, G-box) in 58 *NRT* genes suggests that nitrate transport in potato is tightly regulated by light signals, potentially to synchronize with photosynthetic activity.

MeJA-responsive *cis*-elements were predicted in the promoter regions of 66 *StNRT* genes. As a key plant hormone, jasmonic acid regulates the biosynthesis of secondary metabolites such as terpenes, alkaloids, and flavonoids. Transcription factors binding to these *cis*-elements can activate or inhibit enzymes involved in secondary metabolite biosynthesis, thereby modulating their specific accumulation. This regulatory mechanism highlights the potential role of *StNRT* genes in plant secondary metabolism.

Eighty-one *StNRT* genes tissue specific expression profiling revealed that there were no gene expression data for *StNRT67*. According to the *cis*-elements in the 1 kbs promoter region upstream of *cds*, in addition to the common light-responsive elements, it was found that there was a salicylic acid response, a gibberellin response, a MeJA response, an anaerobic induction, a *cis*-elements involved in defense and stress response, and MYB binding sites involved in drought-induced MYB. The *StNRT67* gene may be associated with potato resistance to abiotic stress.

The promoter region of *StNRT12* harbors 19 *cis*-elements, including those responsive to MeJA, light, abscisic acid, gibberellin, anaerobic induction, and MYB binding sites associated with drought inducibility. The expression of *StNRT12* was positively correlated with nitrogen levels, consistent with the established role of NRT2 proteins in nitrate uptake, which often requires the assistance of NAR2 accessory proteins [45].

Functional insights on StNRT genes

There are multiple nitrate transporters in potato, and there is a considerable degree of functional redundancy. Our analysis revealed distinct expression patterns between the two tandem-duplicated gene pairs. The *StNRT61/62* pair exhibited coordinated transcriptional responses to nitrogen stress, suggesting conserved regulatory mechanisms. In contrast, the *StNRT53/56* pair showed divergent expression patterns under nitrogen stress despite their shared tissue-specific expression profiles. This observation aligns with the well-documented phenomenon of functional redundancy among tandem-duplicated genes [23], where gene pairs may maintain overlapping functions while developing expression divergence.

Notably, while both *StNRT53* and *StNRT56* were transcriptionally active, only *StNRT56* protein was detected at appreciable levels. This discrepancy, coupled with *StNRT53*'s consistently lower transcript abundance, suggests potential subfunctionalization or neofunctionalization following duplication [6]. Such evolutionary outcomes are characteristic of tandem-duplicated genes, where functional redundancy may be gradually resolved through differential expression patterns or protein-level regulation.

Most of the mutants encoding nitrate transporters have no obvious phenotypic differences. Overexpressing the gene *OsNRT1.1A* in rice boosts biomass, yield, and nitrogen use efficiency, especially under low nitrogen conditions, where yields can increase by up to 60%. Similar effects are seen in *Arabidopsis thaliana*, where overexpression accelerates flowering and increases biomass and seed weight. However, mutating *OsNRT1.1A* causes rice plants to become dwarfed, delay flowering, and reduce yield. Additionally, the related gene *AtNRT1.1* in *Arabidopsis* regulates nitrate and auxin transport, influencing lateral root growth [46, 47]. *StNRT38*, *StNRT53*, *StNRT56* (*NPF6.3*) are the homologous gene of *NRT1.1* (Fig. S11). The promoters of these genes contain multiple

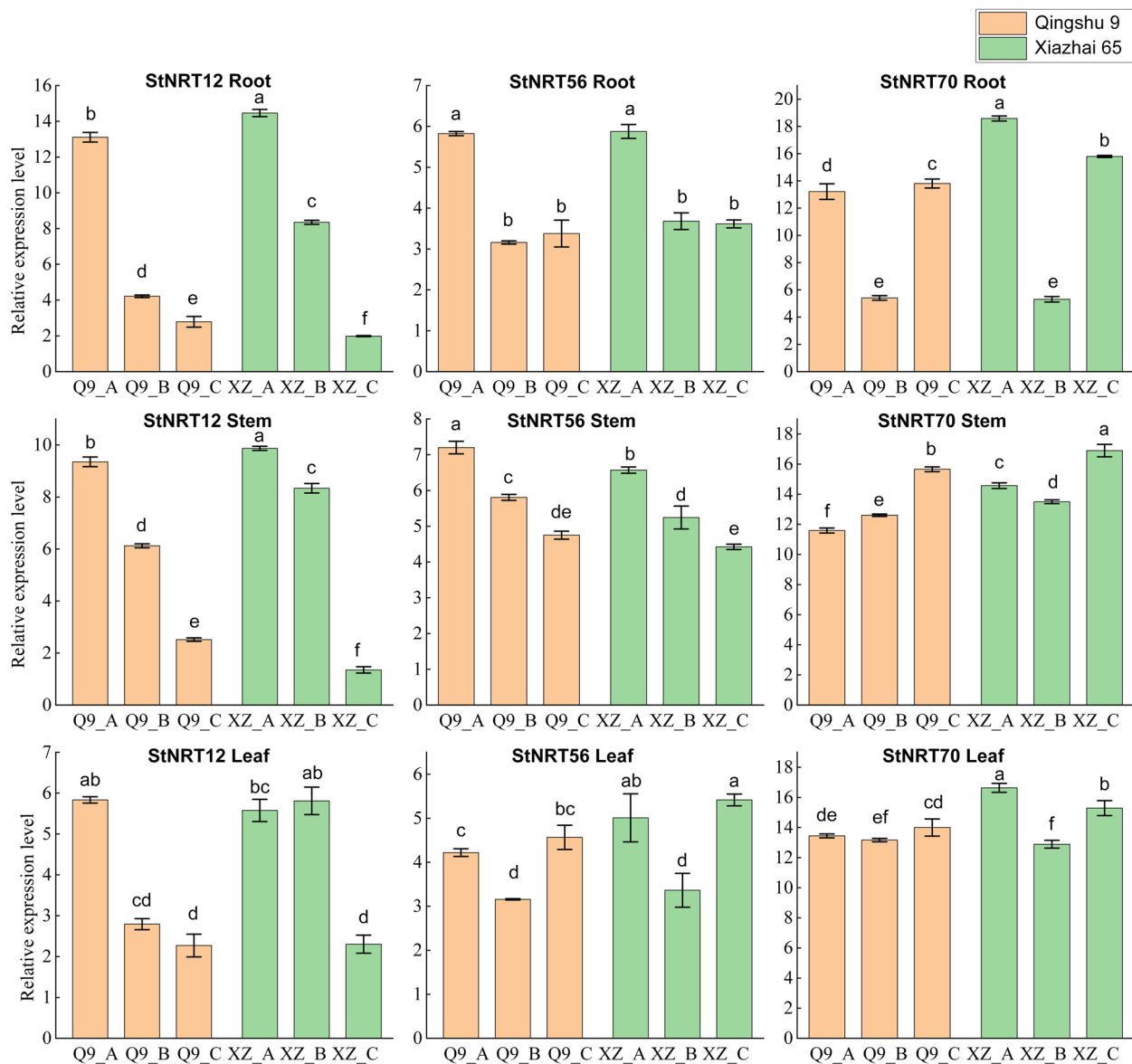


Fig. 9 Expression levels of the StNRT gene in roots, stems and leaves under three nitrogen conditions. Note: **A**, high nitrogen (40 mM NO_3^-); **B**, medium nitrogen (8 mM NO_3^-); **C**, low nitrogen (2 mM NO_3^-); Q9, Qingshu 9; XZ, Xiazhai 65. Significant differences ($p < 0.05$) between treatment groups were indicated by distinct lowercase letters (Tukey's HSD test). Error bars represent $\pm$ SD ($n=3$ biological replicates)

nitrate-responsive cis-elements, e.g. ANR1-binding site, TGA-element, AuxRR-core, ARE, ABRE, TATC-box (see Table S5), which further implicating their regulation by nitrogen signaling. This gene is a functional homologous gene worthy of study in the future.

Evolutionary analysis of NRT gene family

The analysis of Ka/Ks for syntenic gene pairs across multiple species (Table S6) revealed that the majority of gene pairs exhibited Ka/Ks ratios significantly less than 1 ($\text{Ka/Ks} < 1$), indicating that these genes have predominantly undergone purifying selection during evolution. This suggests that mutations within these genes were

generally deleterious to protein function and were consequently eliminated through evolutionary processes. Conversely, a minor subset of gene pairs demonstrated Ka/Ks values approaching or exceeding 1 ($\text{Ka/Ks} \geq 1$), implying potential positive selection, where mutations conferred functional advantages and were thus retained.

Comparative evolutionary analysis between potato (*S. tuberosum*) and tomato (*S. lycopersicum*) revealed consistently low Ka/Ks values (typically < 0.25), indicative of strong evolutionary conservation within the Solanaceae family. In contrast, comparisons between potato and *Arabidopsis thaliana* exhibited higher Ks values (e.g., $\text{Ks} = 2.822$ for AT3G53960.1), reflecting a greater

evolutionary divergence between these species. Similarly, multiple gene pairs in the comparison between potato and rice (*O. sativa*) displayed Ks saturation (e.g., $pS=0.984$ for Os10t0554200-02; pS , Proportion of Synonymous substitutions), suggesting an ancient divergence between monocotyledonous and dicotyledonous plants. Furthermore, when compared to tomato, gene pairs in *Arabidopsis* and rice exhibited significantly higher levels of divergence, underscoring more distant evolutionary relationships among these species. These findings collectively highlight the varying degrees of evolutionary conservation and divergence across the studied species, providing valuable insights into the molecular evolutionary mechanisms shaping their genomes.

Expression of StNRT genes in response to nitrogen stress

Nitrogen stress significantly influenced the expression of *StNRT* genes, particularly those in the *NRT2* family. Under low-nitrogen conditions, *NRT2.1* and *NRT2.2* exhibited rapid upregulation, playing pivotal roles in nitrate uptake [10, 48]. In contrast, *NRT2.4* and *NRT2.5* were induced under chronic nitrogen deficiency [49], with *NRT2.5* emerging as a gene in efficient nitrogen utilization. These findings underscore the critical role of *NRT2* genes in optimizing nitrogen uptake and assimilation under varying nitrogen availability.

Transcription levels were partially consistent with protein expression, with some discrepancies observed. The expression of *StNRT22* exhibited high and consistent levels in both transcription and protein, with tissue-specific expression results also showing alignment, and higher expression levels were detected in all tissues. *StNRT12* gene and M1BJG8 protein displayed consistency, and the expression of the *StNRT35* gene and its corresponding protein was also consistent, with both showing elevated levels in roots. In contrast, *StNRT33* exhibited very low transcriptional expression but demonstrated high protein expression in Xiazhai 65. *StNRT78* was generally detected at higher levels in the transcription of DM8 genome and showed higher consistency in the proteome; however, its transcriptional sequencing content in the Soltub_3.0 genome was lower. These findings suggest that the choice of the reference genome also influences the transcriptional sequencing results of tandem duplication genes. The observed inconsistencies may arise from differences in genetic background, alternative splicing, temporal discrepancies between transcription and translation, post-transcriptional regulatory, epigenetic modifications, mRNA stability, and potential inaccuracies in current gene annotation.

Effective nitrate allocation is essential for plant NUE. NRTs mediate nitrate transport between xylem and phloem, influencing nitrogen remobilization and young tissue development. Interactions (Fig. 10) between

high-affinity NRTs (e.g., StNRT08, 34, 36, 42, 70, and 71) and nitrate reductases (StNR3, M1CYP7) and nitrite reductases (StNIR, M1AEW6 & M1CFI4), suggest a coordinated mechanism for nitrate allocation and utilization. NRT2.4/2.5/2.7 (high-affinity transport systems, HATS) facilitate NO_3^- uptake under low-nitrate conditions in roots. Nitrate Reductase (NR) converts NO_3^- to NO_2^- in the cytoplasm, dependent on NRT-mediated substrate supply. Nitrite Reductase (NiR) further reduces NO_2^- to NH_4^+ in chloroplasts, utilizing photosynthetic-derived ferredoxin. NR and NiR are strongly dependent on NAD(P)H and reduced ferredoxin. GS/GOGAT cycle incorporates NH_4^+ into glutamine/glutamate, serving as nitrogen donors for amino acid and nucleotide synthesis. DHODH (Dihydroorotate Dehydrogenase, quinone, mitochondrial) links nitrogen assimilation to pyrimidine biosynthesis by utilizing glutamine-derived nitrogen. DPD [Dihydropyrimidine Dehydrogenase (NADP (+))] initiates pyrimidine catabolism, enabling nitrogen recycling under stress or senescence [50]. Collectively, these genes collectively participate in the nitrogen metabolism process and stress adaptation in potatoes, particularly in the biological processes of nitrate absorption, transport, reduction, and assimilation into organic nitrogen compounds.

GO enrichment analysis revealed that *StNRT* genes are primarily involved in biological processes such as transmembrane transport and nitrate transport, with localization to the plasma membrane and vacuolar membrane. The presence of cis-elements responsive to light, ABA, and MeJA in the promoter regions suggests that these genes are regulated by hormonal and environmental signals.

Limitations and future directions

This study identified and analyzed nitrate transporters primarily from the NPF (nitrate transporter1/peptide transporter family) and *NRT2* (nitrate transporter2) families in potato. However, two other transporter families — chloride channel (CLC) and slow anion channel homolog (SLAC/SLAH) — were not investigated due to scope constraints. Future studies should include these families to provide a more comprehensive understanding of nitrate transport mechanisms in potato. The transcriptomic and proteomic data reflect the nitrogen response of potato seedlings under in vitro tissue culture conditions, focusing on roots, stems, and leaves at the seedling stage. While the selected varieties (high- and low-nitrogen-efficient) were pre-screened in Qinghai field conditions, the experimental results may not fully represent mature plant responses under natural soil environments. Further validation under field conditions is needed to confirm these findings. Although the SolTub_3.0 genome assembly was used for consistency with prior studies, we

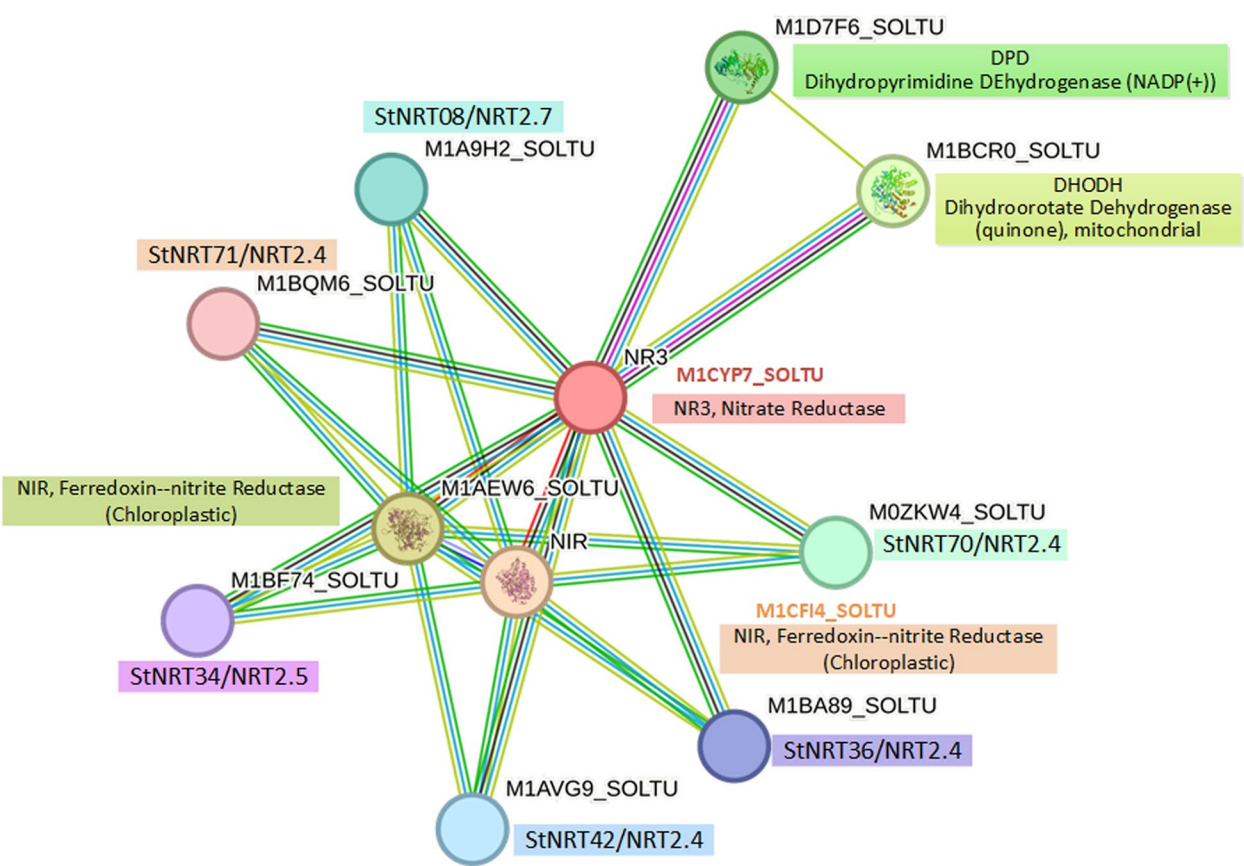


Fig. 10 Gene interactions string. (<https://cn.string-db.org/network/4113.M1CYP7>)

recognize that the latest DM8.1 assembly offers improved telomere resolution. To address this limitation, future work will incorporate long-read sequencing to better characterize tandem duplication events and refine gene family evolutionary analyses. Currently, functional validation of *StNRT* genes remains limited. To bridge this gap, subsequent research will prioritize gene-editing experiments (e.g., CRISPR-Cas9) and genetic transformation to elucidate the roles of key *StNRT*s in nitrate transport and stress adaptation.

Conclusion

Overall, this study provides valuable insights into the functional study of the potato *NRT* gene family. 81 *StNRT* genes were identified comprehensively on genome-wide. Through the study of potato nitrate transporter protein conserved domain, providing a reference for the study of other plant nitrogen transport families and homologous genes. Through the study of potato nitrate transporter expression profiles, understanding how these genes express and integrated function is helpful for identifying the key points that constrain the maximization of potato nitrogen utilization. Based on the mining and functional identification of genes involved in nitrate transport and reuse, future research can employ gene editing and genetic

engineering to alter key gene expression. Thus, modifying these genes in potato varieties are expected to enhance nitrogen transport efficiency and nitrogen utilization.

Abbreviation	
ABA	Absciscic acid
Cds	Coding sequence
CLC	Chlorine channel family transporters
DEGs	Differentially expressed genes
FPKM	Fragments Per Kilo bases per Million fragments
GA ₃	Gibberellin A3
GRAVY	Grand average of hydropathicity
HATs	High-affinity transport systems
HMM	Hidden Markov Model
Ka	non-synonymous substitution rate
Ks	synonymous substitution rate
LATs	Low-affinity transport systems
MFS_1 superfamily	Major Facilitator Superfamily
MFS_NRT2_like	Plant Nitrate transporter NRT2 family and Bacterial Nitrate/Nitrite transporters of the Major Facilitator Superfamily
MFS_NPF1_2	NRT1/PTR family (NPF), subfamily 1 and 2 of the Major Facilitator Superfamily of transporters
MFS_NPF4	NRT1/PTR family (NPF), subfamily 4 of the Major Facilitator Superfamily of transporters
NarK	Nitrate transporter NarK
NAR2	High- affinity nitrate transporter accessory
NUE	Nitrogen use efficiency
pS	Proportion of Synonymous substitutions
T2T	Telomere-to-Telomere
The POT family	The proton-dependent oligopeptide transport family
TMHs	Transmembrane helices
SLAC/SLAH	Slow anion associate channel homolog

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06978-v>.

Additional file 1: Table S1. List of 81 StNRT genes identified in the potato Soltub_3.0 genome. Table S2. List of 96 StNRT genes identified on potato DM8.1 genome and BLASTn result against soltub_3.0 genome. Table S3. Physicochemical properties of 81 StNRT proteins. Table S4. Conserved motif sequences identified in the protein sequences of 81 StNRT genes. Table S5. List of cis-elements identified in the promoter sequences of 81 StNRT genes. Table S6. List of collinear NRT genes between *Solanum tuberosum* and other plants species. Table S7. List of NRT genes identified in tomato and thale cress. Table S8. List of GO and KEGG enriched in StNRT genes. Table S9. Expression values, transcripts per million (TPM) and TAU index for StNRT genes across different potato tissues. Table S10. StNRT Genes Expression values determined via RNA-Seq. Table S11. StNRT protein expression values determined via proteomics. Table S12. Gene-specific primers were used in the RT-qPCR analysis. Table S13. List of 39 StNRT genes identified by Zhang et al. using ST DMv4.03 genome.

Additional file 2: Protein, cds and promoter sequences of NRT/NPFs, 81 StNRT protein sequences, 81 StNRT cds sequences, 81 StNRT cds upstream 1000 bps promoter sequences, 38 AtNRT protein sequences, 85 SINRT protein sequences.

Additional file 3: Fig. S1 Physical map of the StNRT gene on the potato DM8.1 genome. Fig. S2 Conserved domains of 81 StNRT proteins identified by InterPro. Fig. S3 Conserved domains of 96 StNRT proteins identified by InterPro. Fig. S4 Comparison of chromosomal distribution between Zhang's 39 StNRT genes and the 81 StNRT genes. Fig. S5 Conserved domains of 96 StNRT proteins identified using batch CD-Search. Fig. S6 Conserved motifs of 96 StNRT proteins predicted by MEME Suite. Fig. S7 Phylogenetic tree and conserved domains of 96 StNRT proteins identified by InterPro. Fig. S8 Gene structure of 96 StNRT genes. Fig. S9 Predicted cis-elements in the 1kb promoter sequences of 96 StNRT genes. Fig. S10 Phylogenetic tree of StNRTs on DM8.1. Fig. S11 Protein sequences of NRT1.1 homologous genes.

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Authors' contributions

HG, TcN, and JW conceived the study, designed and supervised the experiments; ZxH and WxZ conducted the experiments and analyzed the data; ZxH and JW wrote and revised the manuscript; ZxH and JW contributed equally to this work and should be considered co-first authors; HG and TcN are the corresponding authors. All authors have read and approved the manuscript.

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Data availability

All data generated or analysed during this study are included in this published article and its additional files. The RNA-Seq data are available in the BioProject accession: PRJNA1212866 repository, <https://dataview.ncbi.nlm.nih.gov/object/PRJNA1212866?reviewer=iicmn16hfqdsqdn53mrrm5120a>. The spectrometry proteomics data have been deposited to the ProteomeXchange Consortium with the dataset identifier PXD060191, <http://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PX060191>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing Interests

The authors declare no competing interests.

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References

- Gaudinier A, Rodriguez-Medina J, Zhang L, Olson A, Liseron-Monfils C, Bågman A, Foret J, Abbott S, Tang M, Li B, et al. Transcriptional regulation of nitrogen-associated metabolism and growth. *Nature*. 2018;563(7730):259–64. <https://doi.org/10.1038/s41586-018-0656-3>.
- Xu G, Fan X, Miller AJ. Plant nitrogen assimilation and use efficiency. *Annu Rev Plant Biol*. 2012;63(1):153–82. <https://doi.org/10.1146/annurev-arplant-042811-105532>.
- Bradshaw JE. *Potato breeding: theory and practice*. Switzerland: Springer Nature; 2021. <https://doi.org/10.1007/978-3-030-64414-7>.
- Tang D, Jia Y, Zhang J, Li H, Cheng L, Wang P, Bao Z, Liu Z, Feng S, Zhu X, et al. Genome evolution and diversity of wild and cultivated potatoes. *Nature*. 2022;606(7914):535–41. <https://doi.org/10.1038/s41586-022-04822-x>.
- Guohua M, Fanjun C, Fusuo Z. *Physiological Basis and Genetic Improvement of Nutrient Use Efficiency in Crops*. Beijing: China Agricultural University Press; 2012.
- Wang YY, Cheng YH, Chen KE, Tsay YF. Nitrate transport, signaling, and use efficiency. *Annu Rev Plant Biol*. 2018;69(1):85–122. <https://doi.org/10.1146/annurev-arplant-042817-040056>.
- O'Brien JA, Vega A, Bouguyon E, Krouk G, Gojon A, Coruzzi G, Gutiérrez RA. Nitrate transport, sensing, and responses in plants. *Mol Plant*. 2016;9(6):837–56. <https://doi.org/10.1016/j.molp.2016.05.004>.
- Li J, Fu Y, Pike SM, Bao J, Tian W, Zhang Y, Chen C, Zhang Y, Li H, Huang J, et al. The *Arabidopsis* nitrate transporter NRT1.8 functions in nitrate removal from the xylem sap and mediates cadmium tolerance. *The Plant Cell*. 2010;22(5):1633–46. <https://doi.org/10.1105/tpc.110.075242>.
- Kiba T, Feria-Bourrellier A, Lafouge F, Lezhneva L, Boutet-Mercery S, Orsel M, Bréhaut V, Miller A, Daniel-Vedele F, Sakakibara H, et al. The *Arabidopsis* nitrate transporter NRT2.4 plays a double role in roots and shoots of nitrogen-starved plants. *The Plant Cell*. 2012;24(1):245–58. <https://doi.org/10.1105/tpc.111.092221>.
- Xu N, Cheng L, Kong Y, Chen G, Zhao L, Liu F. Functional analyses of the NRT2 family of nitrate transporters in *Arabidopsis*. *Front Plant Sci*. 2024;15:1351998. <https://doi.org/10.3389/fpls.2024.1351998>.
- Sakuraba Y, Chaganzhana, Mabuchi A, Iba K, Yanagisawa S. Enhanced NRT1.1/NPF6.3 expression in shoots improves growth under nitrogen deficiency stress in *Arabidopsis*. *Commun Biol*. 2021;4(1):256–70. <https://doi.org/10.1038/s42003-021-01775-1>.
- Vidal EA, Alvarez JM, Araus V, Riveras E, Brooks MD, Krouk G, Ruffel S, Lejay L, Crawford NM, Coruzzi GM, et al. Nitrate in 2020: Thirty years from transport to signaling networks. *Plant Cell*. 2020;32(7):2094–119. <https://doi.org/10.1105/tpc.19.00748>.
- Hu B, Jiang Z, Wang W, Qiu Y, Zhang Z, Liu Y, Li A, Gao X, Liu L, Qian Y, et al. Nitrate–NRT1.1B–SPX4 cascade integrates nitrogen and phosphorus signaling networks in plants. *Nat Plants*. 2019;5(4):401–13. <https://doi.org/10.1038/s41477-019-0384-1>.
- Wang W, Hu B, Yuan D, Liu Y, Che R, Hu Y, Ou S, Liu Y, Zhang Z, Wang H, et al. Expression of the nitrate transporter gene *OsNRT1.1A/OsNPF6.3* confers high yield and early maturation in rice. *Plant Cell*. 2018;30(3):638–51. <https://doi.org/10.1105/tpc.17.00809>.

15. Zhang J, Liu Y, Zhang N, Hu B, Jin T, Xu H, Qin Y, Yan P, Zhang X, Guo X, et al. NRT1.1B is associated with root microbiota composition and nitrogen use in field-grown rice. *Nat Biotechnol*. 2019;37(6):676–84. <https://doi.org/10.1038/s41587-019-0104-4>.
16. Hu B, Wang W, Ou S, Tang J, Li H, Che R, Zhang Z, Chai X, Wang H, Wang Y, et al. Variation in NRT1.1B contributes to nitrate-use divergence between rice subspecies *Nat Genet*. 2015;47(7):834–8. <https://doi.org/10.1038/ng.3337>.
17. Yang X, Nong B, Chen C, Wang J, Xia X, Zhang Z, Wei Y, Zeng Y, Feng R, Wu Y, et al. OsNPF3.1, a member of the NRT1/PTR family, increases nitrogen use efficiency and biomass production in rice. *The Crop Journal*. 2023;11(1):108–18. <https://doi.org/10.1016/j.cj.2022.07.001>.
18. Hang J, Wu B, Qiu D, Yang G, Fang Z, Zhang M. *OsNPF3.1*, a nitrate, abscisic acid and gibberellin transporter gene, is essential for rice tillering and nitrogen utilization efficiency. *J Integr Agric*. 2024;23(4):1087–104. <https://doi.org/10.1016/j.jia.2023.04.024>.
19. Luo B, Chen J, Zhu L, Liu S, Li B, Lu H, Ye G, Xu G, Fan X. Overexpression of a high-affinity nitrate transporter *OsNRT2.1* increases yield and manganese accumulation in rice under alternating wet and dry condition. *Front Plant Sci*. 2018;9. <https://doi.org/10.3389/fpls.2018.01192>.
20. Chen J, Qi T, Hu Z, Fan X, Zhu L, Iqbal MF, Yin X, Xu G, Fan X. *OsNAR2.1* positively regulates drought tolerance and grain yield under drought stress conditions in rice. *Front Plant Sci*. 2019; 10:197–209. <https://doi.org/10.3389/fpls.2019.00197>.
21. Shoji T, Saito K. Downregulation of a cluster of genes encoding nitrate transporter 1/peptide transporter family proteins in tomato with a mutated JRE4 transcription factor. *Plant Biotechnol (Tokyo)*. 2023;40(1):71–6. <https://doi.org/10.5511/plantbiotechnology.22.1113a>.
22. Longo A, Miles NW, Dickstein R. Genome mining of plant NPFs reveals varying conservation of signature motifs associated with the mechanism of transport. *Front Plant Sci*. 2018;9:1688–705. <https://doi.org/10.3389/fpls.2018.01668>.
23. Zhang J, Han Z, Lu Y, Zhao Y, Wang Y, Zhang J, Ma H, Han YZ. Genome-wide identification, structural and gene expression analysis of the nitrate transporters (*NRTs*) family in potato (*Solanum tuberosum* L.). *PLoS One*. 2021;16(10). <https://doi.org/10.1371/journal.pone.0257383>.
24. The Potato Genome Sequencing Consortium. Genome sequence and analysis of the tuber crop potato. *Nature*. 2011;475(7355):189–95. <https://doi.org/10.1038/nature10158>.
25. Chen C, Wu Y, Li J, Wang X, Zeng Z, Xu J, Liu Y, Feng J, Chen H, He Y, Xia R. Tbttools-II: a “one for all, all for one” bioinformatics platform for biological big-data mining. *Mol Plant*. 2023;16(11):1733–42. <https://doi.org/10.1016/j.molp.2023.09.010>.
26. Jia L, Hu D, Wang J, Liang Y, Li F, Wang Y, Han Y. Genome-wide identification and functional analysis of nitrate transporter genes (NPF, NRT2 and NRT3) in maize. *Int J Mol Sci*. 2023;24(16):12941. <https://doi.org/10.3390/ijms241612941>.
27. Letunic I, Khedkar S, Bork P. SMART: recent updates, new developments and status in 2020. *Nucleic Acids Res*. 2021;49(D1):D458–60. <https://doi.org/10.1093/nar/gkaa937>.
28. Yang X, Zhang L, Guo X, Xu J, Zhang K, Yang Y, Yang Y, Jian Y, Dong D, Huang S, et al. The gap-free potato genome assembly reveals large tandem gene clusters of agronomical importance in highly repeated genomic regions. *Mol Plant*. 2023;16(2):314–7. <https://doi.org/10.1016/j.molp.2022.12.010>.
29. Chao J, Li Z, Sun Y, Aluko OO, Wu X, Wang Q, Liu G. MG2C: a user-friendly online tool for drawing genetic maps. *Molecular Horticulture*. 2021;1(1):16–9. <https://doi.org/10.1186/s43897-021-00020-x>.
30. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van de Peer Y, Rouzé P, Rombaux S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for *in silico* analysis of promoter sequences. *Nucleic Acids Res*. 2002;30(1):325–7. <https://doi.org/10.1093/nar/30.1.325>.
31. Bailey TL, Johnson J, Grant CE, Noble WS. The MEME Suite. *Nucleic Acids Res*. 2015;43(W1):W39–49. <https://doi.org/10.1093/nar/gkv416>.
32. Wang J, Chitsaz F, Derbyshire MK, Gonzales NR, Gwazdz M, Lu S, Marchler GH, Song JS, Thanki N, Yamashita RA, et al. The conserved domain database in 2023. *Nucleic Acids Res*. 2023;51(D1):D384–8. <https://doi.org/10.1093/nar/gkac1096>.
33. Krogh A, Larsson B, von Heijne G, Sonnhammer ELL. Predicting transmembrane protein topology with a Hidden Markov Model: application to complete genomes. *J Mol Biol*. 2001;305(3):567–80. <https://doi.org/10.1006/jmbi.2000.4315>.
34. Imai K, Nakai K. Tools for the recognition of sorting signals and the prediction of subcellular localization of proteins from their amino acid sequences. *Front Genet*. 2020;11: 607812. <https://doi.org/10.3389/fgene.2020.607812>.
35. The Tomato Genome Sequencing Consortium, Aflitos S, Schijlen E, de Jong H, de Ridder D, Smit S, Finkers R, Wang J, Zhang G, Li N, et al. Exploring genetic variation in the tomato (*Solanum section Lycopersicon*) clade by whole-genome sequencing. *The Plant Journal*. 2014;80(1):136–48. <https://doi.org/10.1111/tpj.12616>.
36. Xu S, Brockmüller T, Navarro-Quezada A, Kuhl H, Gase K, Ling Z, Zhou W, Kreitzer C, Stanke M, Tang H, et al. Wild tobacco genomes reveal the evolution of nicotine biosynthesis. *Proc Natl Acad Sci*. 2017;114(23):6133–8. <https://doi.org/10.1073/pnas.1700073114>.
37. Atwell S, Huang YS, Vilhjálmsson BJ, Willems G, Horton M, Li Y, Meng D, Platt A, Tarone AM, Hu TT, et al. Genome-wide association study of 107 phenotypes in *Arabidopsis thaliana* inbred lines. *Nature*. 2010;465(7298):627–31. <https://doi.org/10.1038/nature08800>.
38. Swarbreck D, Wilks C, Lamesch P, Berardini TZ, Garcia-Hernandez M, Foerster H, Li D, Meyer T, Muller R, Ploetz L, et al. The *Arabidopsis* information resource (TAIR): gene structure and function annotation. *Nucleic Acids Res*. 2007;36(Database issue): D1009–14. <https://doi.org/10.1093/nar/gkm965>.
39. International Rice Genome Sequencing P. The map-based sequence of the rice genome. *Nature*. 2005;436(7052):793–800. <https://doi.org/10.1038/nature03895>.
40. Tamura K, Stecher G, Kumar S. MEGA11: Molecular Evolutionary Genetics Analysis Version 11. *Mol Biol Evol*. 2021;38(7):3022–7. <https://doi.org/10.1093/molbev/msab120>.
41. Letunic I, Bork P. Interactive tree of life (iTOL) v6: recent updates to the phylogenetic tree display and annotation tool. *Nucleic Acids Res*. 2024;52(W1):W78–82. <https://doi.org/10.1093/nar/gkac268>.
42. Hamilton JP, Hansey CN, Whitty BR, Stoffel K, Massa AN, Van Deynze A, De Jong WS, Douches DS, Buell CR. Single nucleotide polymorphism discovery in elite North American potato germplasm. *BMC Genomics*. 2011;12:302. <https://doi.org/10.1186/1471-2164-12-302>.
43. Guo H, Pu X, Jia H, Zhou Y, Ye G, Yang Y, Na T, Wang J. Transcriptome analysis reveals multiple effects of nitrogen accumulation and metabolism in the roots, shoots, and leaves of potato (*Solanum tuberosum* L.). *BMC Plant Biol*. 2022;22(1):282–93. <https://doi.org/10.1186/s12870-022-03652-3>.
44. Feng H, Fan X, Fan X, Liu X, Miller AJ, Xu G. Multiple roles of nitrate transport accessory protein NAR2 in plants. *Plant Signal Behav*. 2011;6(9):1286–9. <https://doi.org/10.4161/psb.6.9.16377>.
45. Kotur Z, Mackenzie N, Ramesh S, Tyerman SD, Kaiser BN, Glass A. Nitrate transport capacity of the *Arabidopsis thaliana* NRT2 family members and their interactions with AtNAR2.1. *New Phytol*. 2012;194(3):724–31. <https://doi.org/10.1111/j.1469-8137.2012.04094.x>.
46. Lay-Pruitt KS, Takahashi H. Integrating N signals and root growth: the role of nitrate transporter NRT1.1 in auxin-mediated lateral root development. *J Exp Bot*. 2020;71(15):4365–8. <https://doi.org/10.1093/jxb/eraa243>.
47. Wang W, Hu B, Li A, Chu C. NRT1.1s in plants: functions beyond nitrate transport. *J Exp Bot*. 2020;71(15):4373–9. <https://doi.org/10.1093/jxb/erz554>.
48. Li M, Wang T, Zhang H, Liu S, Li W, Abou Elwafa SF, Tian H. TaNRT2.1–6B is a dual-affinity nitrate transporter contributing to nitrogen uptake in bread wheat under both nitrogen deficiency and sufficiency. *The Crop Journal*. 2022;10(4):993–1005. <https://doi.org/10.1016/j.cj.2021.11.012>.
49. Zou X, Liu MY, Wu WH, Wang Y. Phosphorylation at Ser28 stabilizes the *Arabidopsis* nitrate transporter NRT2.1 in response to nitrate limitation. *J Integr Plant Biol*. 2020;62(6):865–76. <https://doi.org/10.1111/jipb.12858>.
50. Buchanan BB, Gruissem W, Jones RL. *Biochemistry & molecular biology of plants*. 2nd ed. Wiley-Blackwell; 2015:236,285,696–704.

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